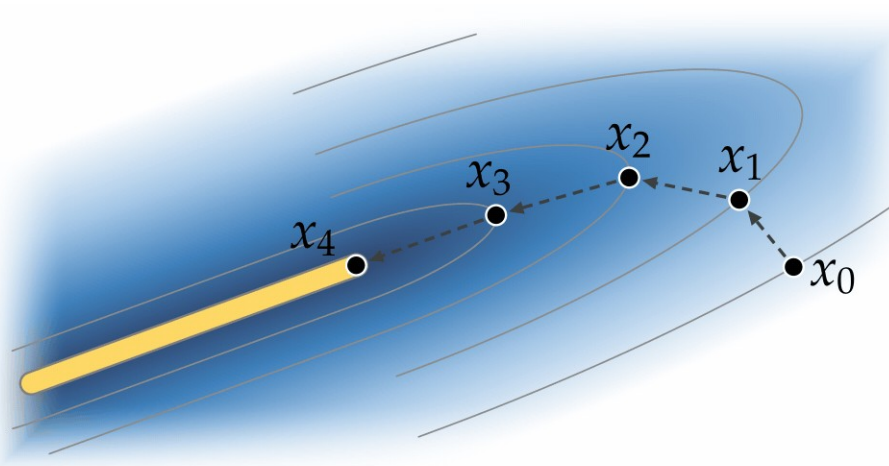


How to Think About Optimization

The Proximal Point Method and Its Impact on Optimization Theory and Machine Learning

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Kingdom of Saudi Arabia



Optimization & Machine Learning Lab

KAUST (King Abdullah University of Science and ...
Oct 23, 2025 • Edited

#KAUST PhD student Kaja Gruntkowska has been awarded a Google PhD Fellowship, earning the 2025 fellowship in Algorithms & Optimization.

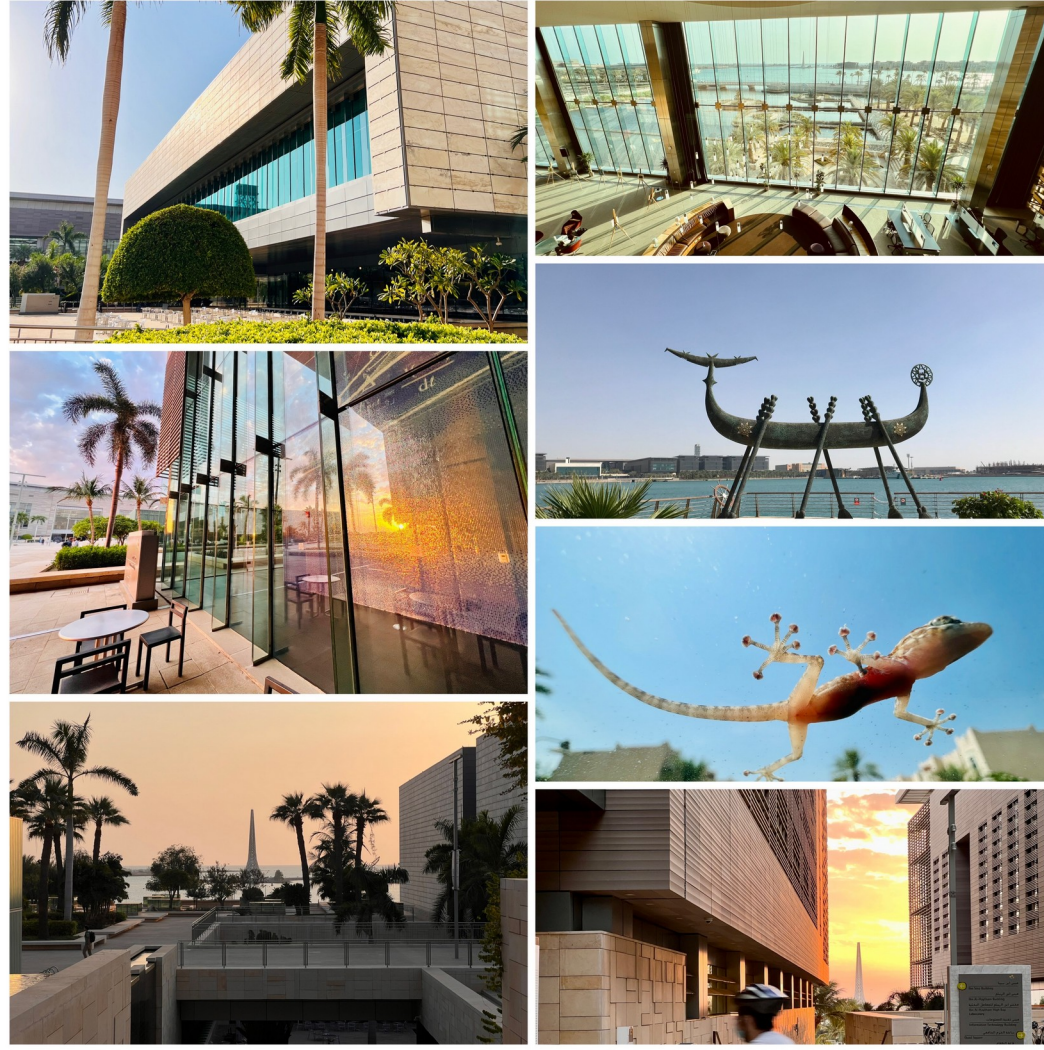
Guided by her advisor, Prof. [Peter Richtarik](#), Kaja's innovative research bridges mathematics, statistics, and AI, advancing both the theory and practice of optimization for machine learning to make AI training faster, more cost-effective, and more resource-efficient.

She shares: "My goal is to bridge rigorous mathematical theory with practical solutions that scale to today's trillion-parameter AI models. This fellowship will help turn these ideas into tools that keep AI breakthroughs within reach."

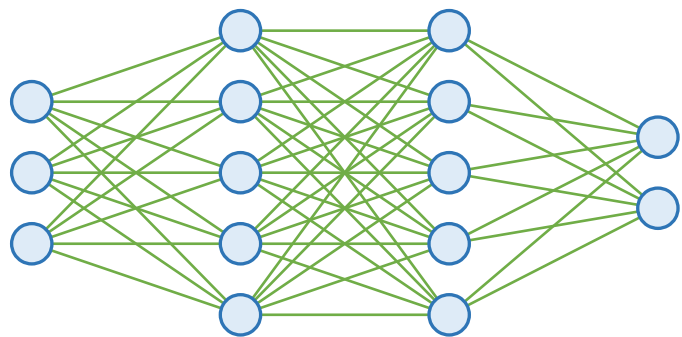
At KAUST, we are proud to empower the next generation of AI leaders and drive innovation from Saudi Arabia to the world.



A Few Random Photos of KAUST



Optimization

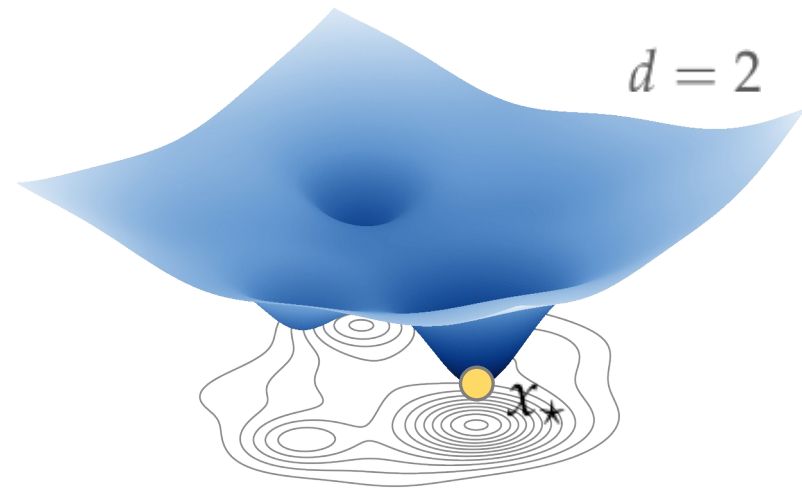


x = weights of a neural network

GPT-3: $d \approx 1.75 \cdot 10^{11}$

$$\min_{x \in \mathbb{R}^d} f(x)$$

number of
trainable
parameters



$$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$$

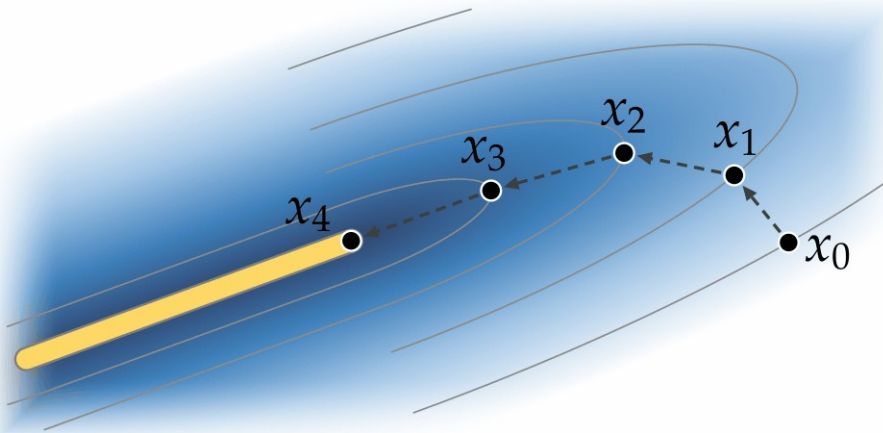
f is proper

f is closed (= lower semi-continuous)

Outline

- Part 1 BPM: Convex Regime
- Part 2 BPM: Convergence Theory
- Part 3 BPM: Nonconvex Regime
- Part 4 BPM: Practical Approximations
- Part 5 BPM: Four Formulations

$$\min_{x \in \mathbb{R}^d} f(x)$$



BPM = Broximal Point Method

arXiv:2003.08078v1 [math.OA] 18 Mar 2020

Acceleration with a Ball Optimization Oracle

Yair Caramon^{*} Arun Jambulapati^{*} Qijia Jiang^{*} Yujia Jin^{*} Yin Tat Lee[†]
Aaron Sidford^{*} Kevin Tian^{*}

Abstract

Consider an oracle which takes a point x and returns the minimizer of a convex function f in an ℓ_2 ball of radius r around x . It is straightforward to show that roughly $r^{-1} \log \frac{1}{\epsilon}$ calls to the oracle suffice to find an ϵ -approximate minimizer of f in an ℓ_2 unit ball. Perhaps surprisingly, this is not optimal: we design an accelerated algorithm which utilizes an ϵ -approximate minimizer with roughly $r^{-1/3} \log \frac{1}{\epsilon}$ oracle queries, and gives a matching lower bound. Further, we implement ball optimization oracles for functions with locally stable Hessians using a variant of Newton's method. The resulting algorithms provide to a number of problems of practical and theoretical import, improving upon previous results for logistic and ℓ_{∞} regression and selecting guarantees comparable to the state-of-the-art for ℓ_2 regression.

1 Introduction

We study unconstrained minimization of a smooth convex objective $f: \mathbb{R}^d \rightarrow \mathbb{R}$, which we access through a ball optimization oracle $\mathcal{O}_{\text{ball}}$ that when queried at any point x , returns the minimizer¹ of f restricted to a ball of radius r around x , i.e.,

$$\mathcal{O}_{\text{ball}}(x) := \arg \min_{y \in B(x, r)} f(y).$$

Such oracles underlie trust region methods [14] and, as we demonstrate via applications, encapsulate problems with local stability. Iteratively $x_{k+1} \leftarrow \mathcal{O}_{\text{ball}}(x_k)$ minimizes f in $\tilde{O}(d/r)$ iterations (see Appendix A), where R is the initial distance to the minimizer, x^* , and $\tilde{O}(\cdot)$ hides polylogarithmic factors in problem parameters, including the desired accuracy.

Given the fundamental geometric nature of the ball optimization abstraction, the central question motivating our work is whether it is possible to improve upon this $\tilde{O}(d/r)$ query complexity. It is natural to conjecture that the answer is negative; we require R/r oracle calls to observe the entire line from x_0 to the optimum, and therefore finding a solution using less queries would require jumping into completely unexplored regions. Nevertheless, we prove that the optimal query complexity scales as $\tilde{O}(d/r)^{1/3}$. This result has positive implications for the complexity for several key regression tasks, for which we can efficiently implement the ball optimization oracles.

1.1 Our contributions

Here we overview the main contributions of our paper: accelerating ball optimization oracles (with a matching lower bound), implementing them under Hessian stability, and applying the resulting techniques to regression problems.

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²In the introduction we discuss exact oracles for simplicity, but our results account for inexactness.

arXiv:2502.02002v1 [math.OA] 1 Feb 2025

The Ball-Proximal (or “Broximal”) Point Method: a New Algorithm, Convergence Theory, and Applications

Kaja Gruntkowska¹ Hammin Li¹ Anil Rao^{1,2} Peter Richtárik¹

Abstract

Non-smooth and non-convex global optimization poses significant challenges across various applications, where standard gradient-based methods often struggle. We propose the *Ball-Proximal Point Method*, *Broximal Point Method*, or *Ball Point Method* (BPM) for short – a novel algorithmic framework inspired by the classical Proximal Point Method (PPM) (Rockafellar, 1976), which, as we show, sheds new light on several foundational optimization paradigms and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. At the core of BPM lies the ball-proximal (“broximal”) operator, which arises from the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the non-smooth convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of ball-convexity, we prove that BPM retains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broad class of practical non-convex optimization problems. Just like PPM plays the role of a conceptual method inspiring the development of practically efficient algorithms and algorithmic elements, e.g., gradient descent, adaptive step sizes, acceleration (Abe & Iwata, 2020), and “W” in Iterative (Zhang et al., 2022), we believe that BPM should be understood in the same manner: as a blueprint and inspiration for further development.

1. **Introduction**
The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing and operations research. Solving such problems is integral to most machine learning algorithms arising in both training and inference, where non-convex objectives or constraints are often necessary to capture complex prediction tasks.

1.1. Global non-convex optimization

In this paper, we propose a new meta-algorithm (see Section 1.3) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (potentially non-convex) optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is assumed to be proper (which means that the set $\text{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and have at least one minimizer. We let X_f be the set of all minimizers of f , and $f_* := \min_x f(x)$.

1.2. The ball-proximal operator

A key inspiration for our method stems from the well-known Proximal Point Method (PPM) (Rockafellar, 1976), which iteratively adds a quadratic penalty term to the objective (see Section 3 for more details) and solves a modified subproblem at each step. Building on this idea, we introduce the ball-proximal (“broximal”) operator.

Definition 1.1 (Ball-Proximal Operator). The ball-proximal (“broximal”) operator with radius $r > 0$ associated with a function $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\text{brox}_r(f)(x) := \arg \min_{y \in B(x, r)} f(y), \quad (2)$$

where $B(x) := \{z \in \mathbb{R}^d : \|x - z\| \leq 1\}$ is the standard Euclidean norm.

According to this definition, for a given input point $x \in \mathbb{R}^d$, $\text{brox}_r(f)$ returns the minimizer of f within the ball of radius r centered at x .

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²The work of Anil Rao was conducted during a VSRP internship at KAUST.

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NON-EUCLIDEAN BROXIMAL POINT METHOD: A BLUEPRINT FOR GEOMETRY-AWARE OPTIMIZATION

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Abstract

The recently proposed Broximal Point Method (BPM) (Gruntkowska et al., 2025) offers an idealized optimization framework based on iteratively minimizing the objective function over norm balls centered at the current iterate. It enjoys striking global convergence guarantees, converging linearly and in a finite number of steps for proper, closed and convex functions. However, its theoretical analysis has so far been confined to the Euclidean geometry. At the same time, emerging trends in deep learning optimization, exemplified by algorithms such as Moen (Jordan et al., 2024) and Stein (Petrick et al., 2025), demonstrate the practical advantages of minimizing over balls defined via non-Euclidean norms which better align with the underlying geometry of the associated loss landscapes. In this note, we ask whether the convergence theory of BPM can be extended to this more general, non-Euclidean setting. We give a positive answer, showing that most of the elegant guarantees of the original method carry over to arbitrary norm geometries. Along the way, we clarify which properties are preserved and which necessarily break down when leaving the Euclidean realm. Our analysis positions Non-Euclidean BPM as a conceptual blueprint for understanding a broad class of geometry-aware optimization algorithms, shedding light on the principles behind their practical effectiveness.

1 Introduction

Optimization stands as a cornerstone of modern machine learning, powering the success of virtually everything, from deep neural networks to state-of-the-art large language models. As models grow ever larger and more complex, the demand for better optimization algorithms evolves in tandem. What is needed now is a new class of methods—one that are not only computationally efficient and scalable, but also inherently robust to the nonconvex, high-dimensional landscapes that characterize contemporary deep learning.

The field of deep learning optimization has long relied on Adam and related algorithms (Kingma and Ba, 2014; Loshchilov and Hutter, 2017), whose core innovation lies in adaptive moment estimation. While these methods have achieved significant empirical success, their theoretical behavior, particularly in nonconvex landscapes, remains only partially understood. Recently, however, a new class of optimization methods has begun to challenge this long-standing dominance. Algorithms such as Moen (Jordan et al., 2024), Stein (Petrick et al., 2025) and Quon (Fridman et al., 2025) break away from the adaptive moment paradigm. Instead, they adopt a different principle: iterated updates derived by minimizing a linear approximation of the loss function over a carefully chosen norm ball. Crucially, these norm balls are non-Euclidean, aiming to reflect the intrinsic geometry of the optimization problem (see Section 3.1). Operating at the individual layer level, these algorithms combine simplicity, scalability, and promising empirical performance, with several studies suggesting their potential to supersede AdamW in large-scale deep learning tasks (Jin et al., 2025; Petrick et al., 2025; Shao et al., 2025; Tsvet et al., 2025).

While Moen and Stein—both of which utilize spectral norm balls—are among the most prominent and actively studied examples in this emerging line of work, the underlying principle extends well beyond these specific instances. By changing the geometry through altering the norms, one can recover a range of familiar optimization methods. For

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Broximal Alignment for Global Non-Convex Optimization

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Abstract

Most non-convex optimization theory is built around gradient dynamics, leaving global convergence largely unexplored. The dominant paradigm focuses on stationarity, certifying only that the gradient norm vanishes, which is often a weak proxy for actual optimization success. In practice, gradient norms can stagnate or even increase during training, and stationary points may be far from global solutions. In this work, we propose a new framework for global non-convex optimization that avoids gradient-based reasoning altogether. We revisit the Ball-Proximal Point Method (BPM), a trust-region-style algorithm introduced by Gruntkowska et al. (2025), and propose a novel structural condition – *Broximal Alignment* – under which BPM provably converges to a global minimizer. Our condition requires convexity, smoothness, or Lipschitz assumptions, and it permits multiple and disconnected global minima as well as non-optimal local minima. We show that this class generalizes standard non-convex frameworks such as quasi-convexity, star-convexity, quasi-convexity, and the aiming condition. Our results provide a new conceptual foundation for the global non-convex optimization beyond stationarity.

deriving landscape can be highly irregular, with many poor local minima and saddle points that can trap optimization algorithms (Pucciani et al., 2014; Vidal et al., 2017; Saffari & Shamsi, 2018; Jin et al., 2021).

Unlike convex optimization, where global minimizers can be efficiently found (Nemirovski & Yudin, 1983; Nemirovski & Nesterov, 1985; Nesterov, 2003), we lack a comparable toolkit for non-convex problems. Indeed, many non-convex optimization tasks are NP-hard to solve (Moré & Rabat, 1987), and a number of them are NP-hard even to approximate (Meka et al., 2008; Jain & Kar, 2017). As a result, rather than aiming for (approximate) global solutions, most theoretical results for non-convex optimization, especially those for gradient-based methods, are framed in terms of stationarity: convergence to points where the gradient norm is small. Although mathematically convenient, such guarantees have brought limited insight into global optimization behavior.

1.1 Limitations of Stationarity-Based Non-Convex Optimization Theory

To fix notation, we consider solving the generic (potentially non-convex) optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper (that is, the set $\text{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and has at least one minimizer. We denote the set of all minimizers by X_f and write $f_* := \min_x f(x)$.

The dominant theoretical paradigm for such tasks is built around stationarity. Convergence guarantees are typically stated in terms of the norm of the gradient, asserting that an algorithm reaches a point where $\|\nabla f(x)\|$ is small. This framework underlies much of the modern theory of stochastic gradient methods.

From the modeling perspective, however, stationarity-based guarantees are fundamentally misaligned with the goals of learning. A stationary point can be a global minimizer, a poor local minimizer, or a saddle point; first-order information alone does not distinguish among these possibilities. Therefore, a small gradient provides little guidance on whether an algorithm meaningfully solves the under-

Preprint. April 16, 2026.

A Sharper Theory of Ball-Proximal Optimization: Convergence and Radius Selection

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28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration picks the objective over a ball centered at the current point. Retaining the algorithm up to the distance of squared distance yields sharper bounds on objective values, stationarity, and the symmetric Bregman distance to a minimizer. For constant radius $r > 0$ and initial distance $D_0 > 0$ to the solution set, the objective gap after k iterations is at most its initial value multiplied by $\exp(-k/r^2 D_0^2)$.

We characterize convergence for arbitrary positive radius sequences. If their sum diverges, the method reaches a minimizer in finitely many iterations whenever one enters otherwise, the objective values converge to the infimum and the iterates escape every bounded set. If the radii are summable, the iterates converge to a possibly nonoptimal point. Self-contraction gives finite trajectory length for bounded iterates and a constant-radius termination bound $O(k^2 + D_0/r)$, whose implicit constant depends only on the dimension. A polished family shows that the dimension-independent quadratic bound remains asymptotically sharp.

We also identify a uniform nonvacuous geometric decay factor and analyze adaptive radius rules based on subgradients or objective gaps, eigengap reformulation, and related updates. Together, these results strengthen the foundation and convergence guarantees of the method without assuming smoothness or strong convexity.

Keywords: ball-proximal point method; convex optimization; finite termination; self-contracted sequences; radius selection; oracle complexity.
Mathematics Subject Classification (2020): Primary 90C25; Secondary 90C60, 65R05.

1 Introduction

We study the convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume neither f nor the infimum is attained nor that f is smooth. We focus on the exact ball-oracle iteration considered by Caramon et al. (2020, Appendix A) and subsequently developed as the ball-proximal (broximal) point framework by Gruntkowska et al. (2025). Starting from $x^0 \in \text{dom} f$, the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point:

$$x^{k+1} \in \arg \min_{x \in B(x^k, r)} f(x), \quad t_k > 0, \quad (\text{BPM})$$

Part 1

Proximal Point Method: Convex Regime



Kaja Gruntkowska, Hanmin Li, Aadi Rane, P.R.

The ball-proximal (= "proximal") point method: a new algorithm, convergence theory, and applications

arXiv:2502.02002, 2025



Kaja Gruntkowska, P.R.

Non-Euclidean proximal point method: a blueprint for geometry-aware optimization

arXiv:2510.00823, 2025

BPM with ℓ_2 Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM run

$$d_k = \operatorname{dist}_2(x_k, \operatorname{Argmin} f)$$

$$f_k = f(x_k) - f_\star$$

$$g_k = \|\nabla f(x_k)\|_2$$

$$d_0 = 5.06 \quad f_0 = 35.78 \quad g_0 = 28.40$$

$$t_0 = 1.0 \quad d_1 = 4.50 \quad f_1 = 14.97 \quad g_1 = 13.61$$

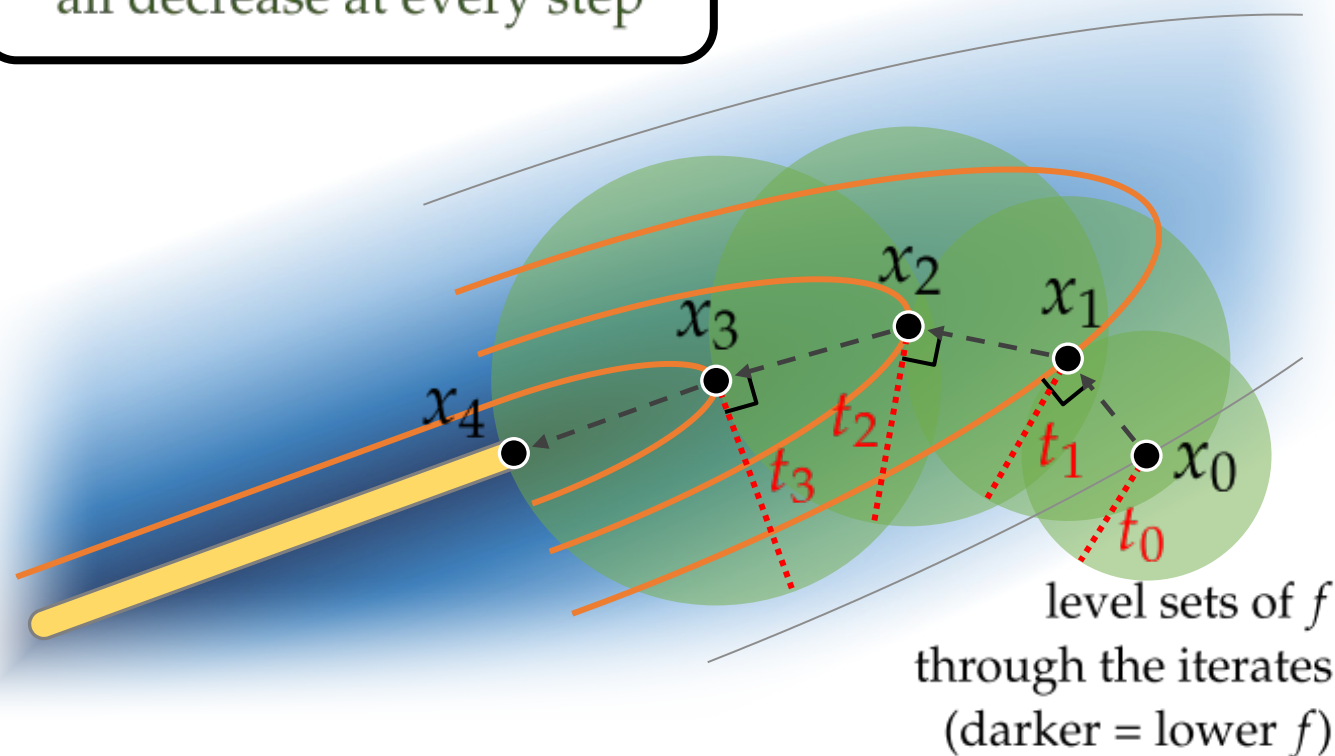
$$t_1 = 1.3 \quad d_2 = 3.32 \quad f_2 = 5.62 \quad g_2 = 3.89$$

$$t_2 = 1.6 \quad d_3 = 1.72 \quad f_3 = 1.48 \quad g_3 = 1.72$$

$$t_3 = 1.8 \quad d_4 = 0.00 \quad f_4 = 0.00 \quad g_4 = 0.00$$

d_k, f_k and g_k
all decrease at every step

$$\operatorname{Argmin} f = [-2a, 2a]$$
$$f_\star := \min f = 0$$



Proximal Point vs. Broximal Point

PPM: Proximal Point Method (Martinet, 1970)

$$x_{k+1} = \operatorname{argmin}_y \left\{ f(y) + \frac{1}{2\gamma} \|y - x_k\|_2^2 \right\}$$

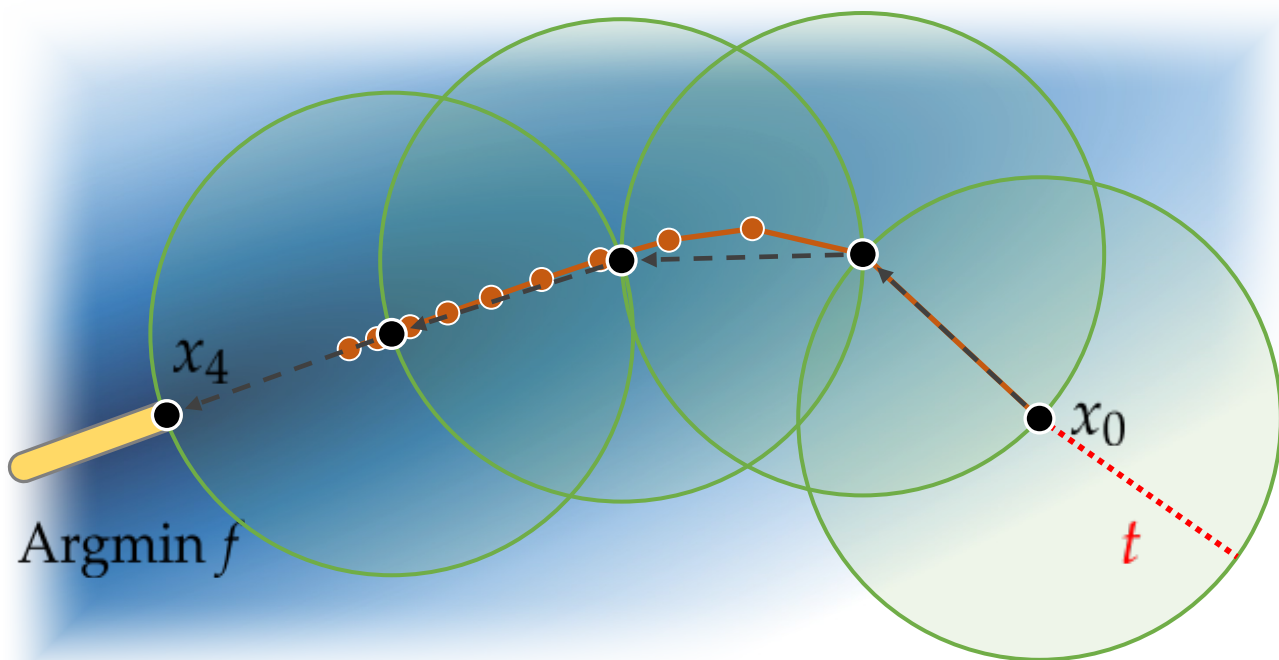
$$\|x_{k+1} - x_k\|_2 = \gamma \|\nabla f(x_{k+1})\|_2 \rightarrow 0$$

f differentiable: $x_k \notin \text{Argmin } f$ for every k

BPM: Broximal (= ball-proximal) Point Method

$$x_{k+1} \in \underset{y}{\operatorname{Argmin}} \left\{ f(y) : \|y - x_k\|_2 \leq t \right\}$$

$\|x_{k+1} - x_k\|_2 = t$ until the ball meets $\text{Argmin } f$
 exact after $\lceil d_0^2/t^2 \rceil$ steps; no condition number



- PPM, $\gamma = 0.155$
- BPM, $t = 1.4$

same x_0 , same first step

after 4 steps: $d_4 = 0$ vs. $d_4 = 2.67$

$$d_k := \text{dist}_2(x_k, \text{Argmin } f)$$

Penalty $\rightarrow$ constraint: the steps
no longer shrink, so BPM reaches
 $\text{Argmin } f$ in finitely many steps.

BPM with ℓ_1 Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_1 \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM run

$$d_k = \operatorname{dist}_1(x_k, \operatorname{Argmin} f)$$

$$f_k = f(x_k) - f_\star$$

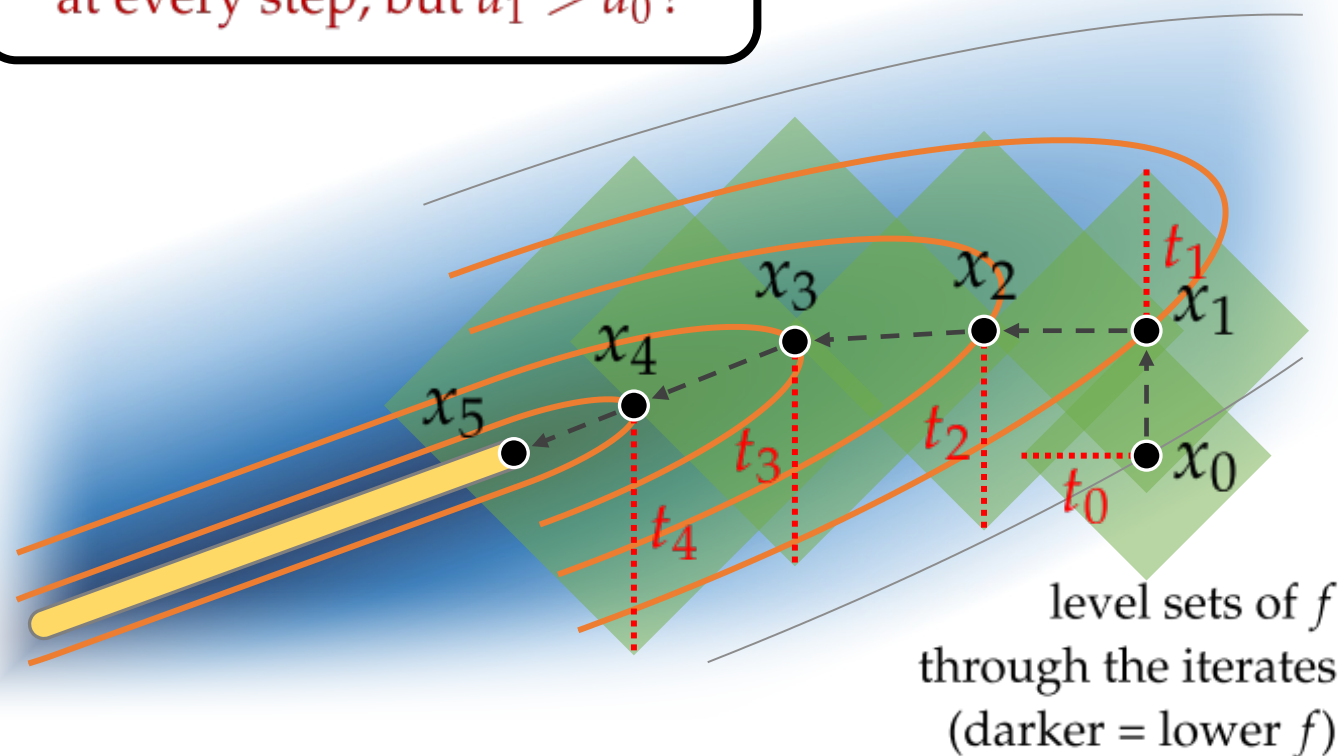
$$g_k = \|\nabla f(x_k)\|_\infty \quad (\text{dual norm})$$

	$d_0 = 5.08$	$f_0 = 35.78$	$g_0 = 24.69$
$t_0 = 1.0$	$d_1 = 6.04 \uparrow$	$f_1 = 18.22$	$g_1 = 10.44$
$t_1 = 1.3$	$d_2 = 4.74$	$f_2 = 8.56$	$g_2 = 5.64$
$t_2 = 1.6$	$d_3 = 3.14$	$f_3 = 2.96$	$g_3 = 1.89$
$t_3 = 1.8$	$d_4 = 1.34$	$f_4 = 0.54$	$g_4 = 0.81$
$t_4 = 2.0$	$d_5 = 0.00$	$f_5 = 0.00$	$g_5 = 0.00$

f_k and g_k decrease
at every step, but $d_1 > d_0$!

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$



BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_\infty \leq t_k\}$$

Example: $f(x) = \frac{1}{2}(|\langle a, x \rangle| - 2)_+^2 + 8\langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = f(x_k) - f_\star$$

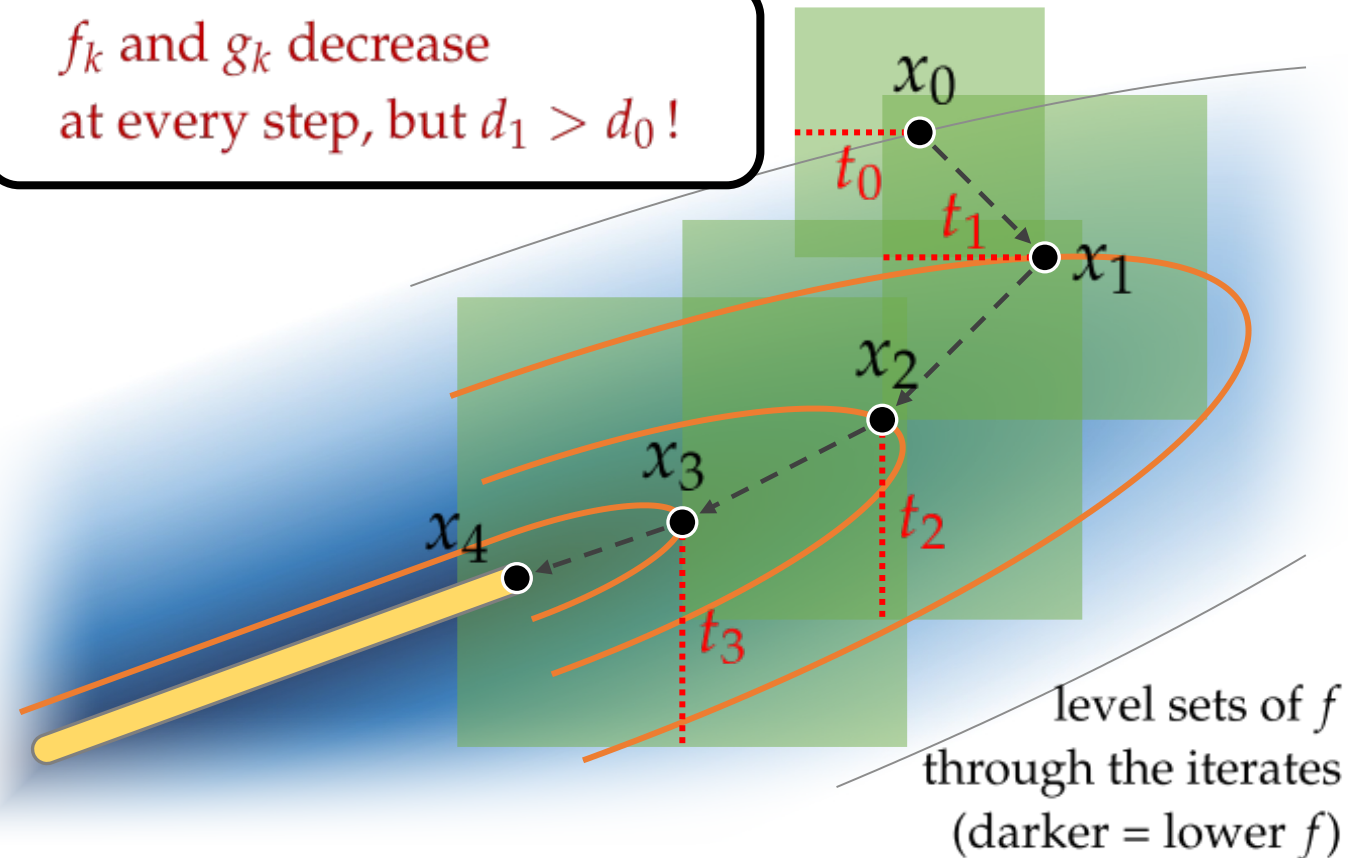
$$g_k = \|\nabla f(x_k)\|_1 \quad (\text{dual norm})$$

	$d_0 = 3.57$	$f_0 = 49.53$	$g_0 = 43.60$
$t_0 = 1.0$	$d_1 = 4.22 \uparrow$	$f_1 = 19.25$	$g_1 = 16.96$
$t_1 = 1.3$	$d_2 = 2.92$	$f_2 = 5.35$	$g_2 = 5.91$
$t_2 = 1.6$	$d_3 = 1.32$	$f_3 = 0.98$	$g_3 = 1.49$
$t_3 = 1.8$	$d_4 = 0.00$	$f_4 = 0.00$	$g_4 = 0.00$

$$\operatorname{Argmin} f = [-2a, 2a]$$

$$f_\star := \min f = 0$$

f_k and g_k decrease
at every step, but $d_1 > d_0$!



Linear Programming via BPM with ℓ_∞ Balls

$$x_{k+1} \in \operatorname{Argmin} \{ \langle c, y \rangle : y \in P, \|y - x_k\|_\infty \leq t \}$$

$$= \operatorname{Argmin} \{ \langle c, y \rangle : \langle a_i, y \rangle \leq b_i \ \forall i \in \mathcal{I}_k, \|y - x_k\|_\infty \leq t \}$$

Example: $f(x) = \langle c, x \rangle + \delta_P(x)$, $P = \{x \in \mathbb{R}^2 : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\}$, $m = 12$, $c = (1, 2)$, $t = 1.4$

BPM run

$$d_k = \operatorname{dist}_\infty(x_k, \operatorname{Argmin} f)$$

$$f_k = \langle c, x_k \rangle - f_\star$$

$$\mathcal{I}_k = \{i : b_i - \langle a_i, x_k \rangle < t \|a_i\|_1\}$$

$$d_0 = 5.05 \quad f_0 = 11.45 \quad \mathcal{I}_0 = \emptyset$$

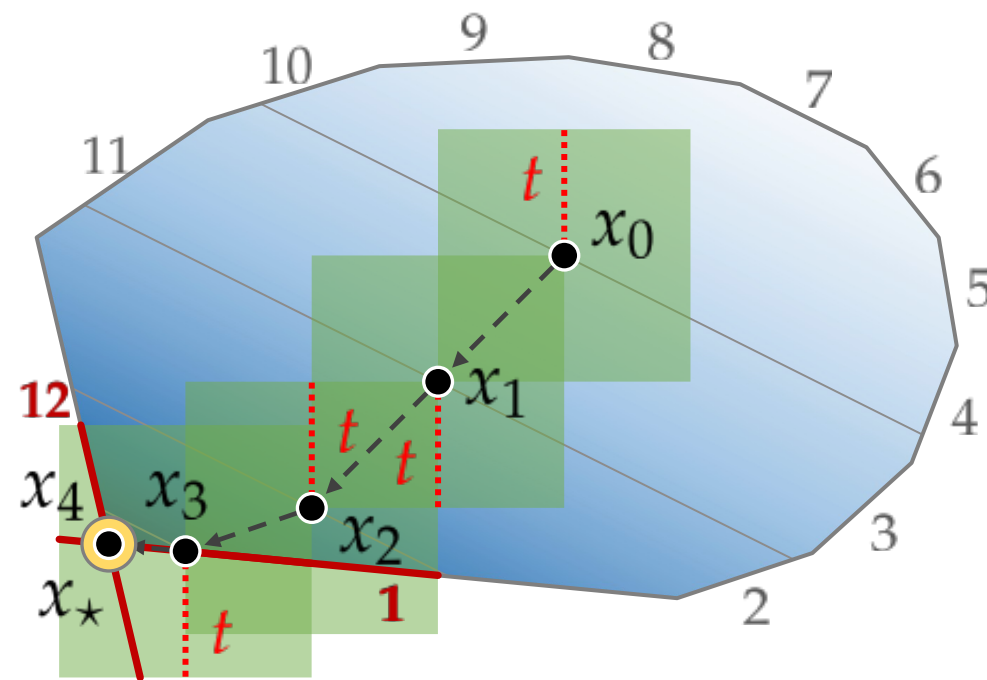
$$d_1 = 3.65 \quad f_1 = 7.25 \quad \mathcal{I}_1 = \emptyset$$

$$d_2 = 2.25 \quad f_2 = 3.05 \quad \mathcal{I}_2 = \{1\}$$

$$d_3 = 0.85 \quad f_3 = 0.69 \quad \mathcal{I}_3 = \{1, 12\}$$

$$d_4 = 0.00 \quad f_4 = 0.00 \quad x_4 = x_\star$$

level sets of f
(darker = lower f)



P.R.

Rethinking Gradient Descent for Linear Programming: A Screened Broximal Approach
Manuscript, 2026

LP $\longrightarrow$ 4 small LPs, one per BPM step;
each uses only $|\mathcal{I}_k| \leq 2$ of the $m = 12$ constraints

Part 2

Broximal Point Method: Convergence Theory



Kaja Grunkowska, Hanmin Li, Aadi Rane, P.R.

The ball-proximal (= "broximal") point method: a new algorithm, convergence theory, and applications

arXiv:2502.02002, 2025



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

Problem Formulation & Assumptions

$$\min_{x \in \mathbb{R}^d} f(x)$$

$$x_{k+1} \in \operatorname{Argmin} \{f(y) : \|y - x_k\|_2 \leq t_k\}$$

$f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed & convex

$$x_\star \in \operatorname{Argmin} f \neq \emptyset, \quad f_\star := \min f$$

$$f(x_0) \neq +\infty$$

arXiv:2502.0202v1 [math.OA] 4 Feb 2025

The Ball-Proximal ("Brossimal") Point Method:
a New Algorithm, Convergence Theory, and Applications

Kaj Gerdikova¹ · Hanmin Li¹ · Adi Ros^{1,2} · Peter Richtárik¹

Abstract

Non-convex and non-smooth global optimization poses significant challenges across various applications, where finding global minima is often intractable. We propose the *Ball-Proximal Point Method*, *Proximal Point Method*, or *Ball-Prox Method* (BPM) for short – a novel algorithm inspired by the classical Proximal Point Method (PPM) [Rockafellar, 1976], which, as we show, sheds new light on several fundamental optimization principles and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. As the core of BPM is the ball-proximal "Proximal" operator, which iterates the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the non-convex convex regime, we prove that BPM converges linearly and in a finite number of steps in the same regime. Furthermore, by introducing the concept of *ball-convexity*, we prove that BPM attains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broader class of practically non-convex optimization problems. Our key BPM gives the birth of a conceptual method inspiring the development of practically efficient algorithms and algorithmic directions, e.g., parallel descent, adaptive step sizes, acceleration (AR & SA, 2020), and "PP" as listed (Zhang et al., 2023), we believe that BPM should be understood as the same manner as the proximal and subgradients for further developments.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing, and operations research. Finding such problems is integral to most machine learning algorithms aiming to both training and inference, where non-convex objectives or constraints are often necessary to capture complex patterns in data.

1.1. Global non-convex optimization

In this paper, we propose a new *non-convex algorithm* (see Section 1.1) capable of finding global minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (generally) non-convex optimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \quad (1)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is bounded below and which means that the set $\text{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty, closed, and has a finite non-emptiness. We let A_f be the set of all minimizers of f and $f_\star := \min_x f(x)$.

1.1.1. The ball-proximal operator

Also inspired by our method stems from the well-known Proximal Point Method (PPM) [Rockafellar, 1976], which iteratively adds a quadratic penalty term to the objective and solves. For non-convex and non-smooth optimization problems at each step. Building on this idea, we introduce the ball-proximal "Proximal" operator

Definition 1.1 (Ball-Proximal Operator). The ball-proximal ("Proximal") operator with radius $r > 0$ associated with a function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is given by

$$\text{ball-prox}_r f(x) := \arg\min_{y \in B(x, r)} f(y), \quad (2)$$

where $B(x, r) := \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}$ is the standard Euclidean ball.

According to this definition, for a given point $x \in \mathbb{R}^d$, $\text{ball-prox}_r f(x)$ returns the minimizers of f within the ball of radius r centered at x .

A Sharper Theory of Ball-Proximal Optimization:
Convergence and Radius Selection

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28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Assuming the objective f is the minimum of squared distance (quadratic) terms on a separable values, elements, and the constraint is Bregman distance to a reference. For convex radius $r > 0$ and initial distance d_0 , the algorithm attains the sharp rate after k iterations is at most its initial value multiplied by $\exp(-k/r^2/d_0)$.

We also investigate convergence for arbitrary points and radius selection. If the ball-proximal operator converges to a globally unique point, ball-convexity gives finite trajectory length for bounded iterates and a constant radius comparison bound $d_0 \leq 2/r$. For $r > 2/r$, more explicit constant depending on the dimension d is provided, which shows that the algorithm's trajectory length is bounded by a constant independent of d .

We also identify a minimum normed trajectory lower bound and analyze why other such bound on algorithmic stepsize, stepsize estimation, and radius selection. Together, these results strengthen the foundations and convergence guarantees of the method without assuming smoothness or strong convexity.

Keywords: ball-proximal point method, convex optimization, finite termination, self-contracted sequences, radius selection, convex conjugate.

Mathematics Subject Classification (2010): Primary 90C30; Secondary 90C06, 49J55.

1 Introduction

We study the convex optimization problem

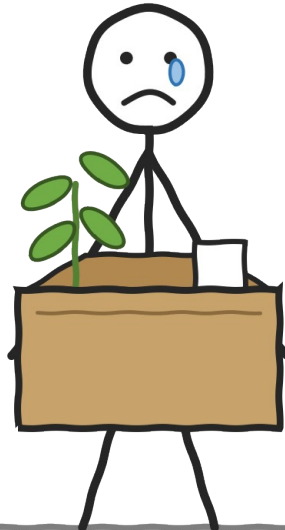
$$\min_{x \in \mathbb{R}^d} f(x), \quad (3)$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume further that the following is satisfied: we let $f_\star := \min_x f(x)$. We focus on the exact ball-proximal iteration considered by Csiszar et al. (2020, Appendix A) and subsequently developed as the ball-proximal (biprox) point framework by Gerdikova et al. (2023). Starting from x^0 , then f , the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point

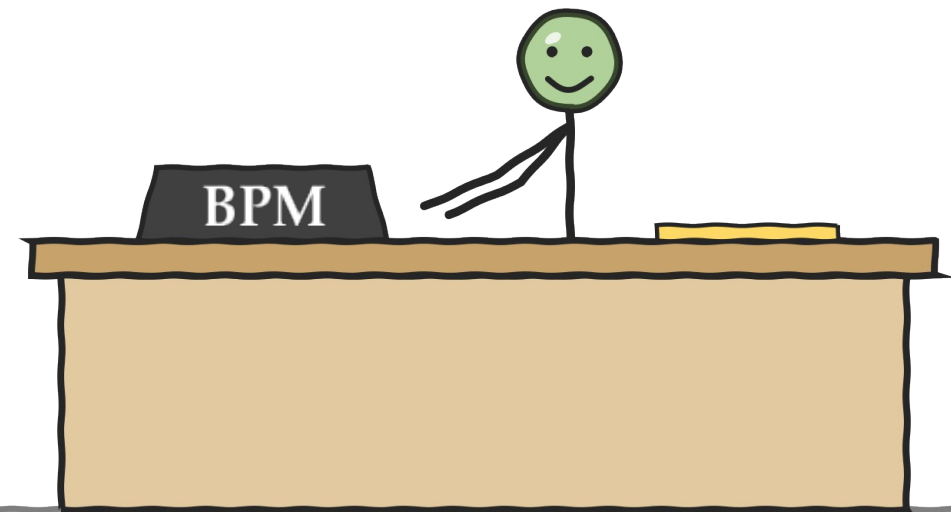
$$x^{k+1} \in \arg\min_{y \in B(x^k, r)} f(y), \quad \text{for } k \geq 0. \quad (BPM)$$

The Condition Number

You can't just leave me out.
I've been in every convergence bound
for fifty years.



We've decided to go
in a different direction.



Fixed-Radius Theory: Distance

$$d_k^2 \leq \max \left\{ d_0^2 - k t^2, 0 \right\}$$

$$d_k = \text{dist}_2(x_k, \text{Argmin } f)$$

$$f_k = f(x_k) - f_*$$

$$g_k = \text{dist}_2(0, \partial f(x_k))$$

$$k \geq \frac{d_0^2}{t^2} \implies d_k = 0$$

No dependence
on a condition
number!



Kaja Grunkowska, Hanmin Li, Aadi Rane, P.R.

**The ball-proximal (= "broximal") point method: a new algorithm,
convergence theory, and applications**

arXiv:2502.02002, 2025

Fixed-Radius Theory: Function

$$f_k \leq f_0 \cdot e^{-2k^2 t^2 / d_0^2}$$

$$d_k = \text{dist}_2(x_k, \text{Argmin } f)$$

$$f_k = f(x_k) - f_*$$

$$g_k = \text{dist}_2(0, \partial f(x_k))$$

$$k \geq \frac{d_0}{\sqrt{2} t} \sqrt{\ln \frac{f_0}{\varepsilon}} \implies f_k \leq \varepsilon$$

No dependence
on a condition
number!



P.R., Hanmin Li

A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

Fixed-Radius Theory: Gradient

$$g_{k+1} \leq g_k$$

$$g_k \leq \frac{f_0}{t} \cdot e^{-2(k-1)^2 t^2 / d_0^2}$$

$$d_k = \text{dist}_2(x_k, \text{Argmin } f)$$

$$f_k = f(x_k) - f_*$$

$$g_k = \text{dist}_2(0, \partial f(x_k))$$

$$k \geq 1 + \frac{d_0}{\sqrt{2} t} \sqrt{\ln \frac{f_0}{t \varepsilon}} \implies g_k \leq \varepsilon$$

No dependence
on a condition
number!



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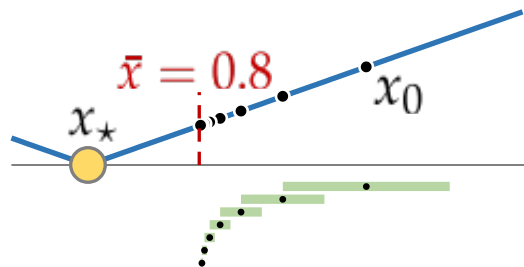
Arbitrary-Radius Theory: Convergence

Theorem. f proper, closed, convex, $x_0 \in \text{dom } f$, any radii $t_k > 0$. Exactly one case applies:

(i) $\sum_k t_k < \infty$:

$x_k \rightarrow \bar{x}$, $f(x_k) \downarrow f(\bar{x})$

$\bar{x}$ need *not* be a minimizer



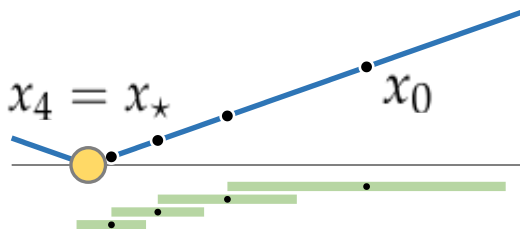
$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = 0.6 \cdot 2^{-k} \Rightarrow x_k \rightarrow 0.8$$

(ii) $\sum_k t_k = \infty$, $\text{Argmin } f \neq \emptyset$:

$x_K \in \text{Argmin } f$ for some $K < \infty$

finite termination



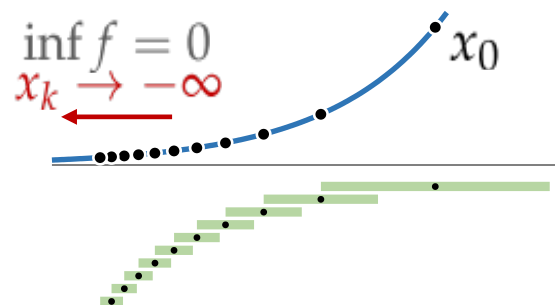
$$f(x) = |x|, \quad x_0 = 2$$

$$t_k = \frac{1}{k+1} \Rightarrow x_4 = 0$$

(iii) $\sum_k t_k = \infty$, $\text{Argmin } f = \emptyset$:

$f(x_k) \downarrow \inf f$, $\|x_k\|_2 \rightarrow \infty$

also when $\inf f = -\infty$



$$f(x) = e^x, \quad x_0 = 1$$

$$t_k = \frac{1}{k+1} \Rightarrow x_k \rightarrow -\infty$$

In particular, $\sum_k t_k = \infty \implies f(x_k) \downarrow \inf f$; radii may shrink, e.g. $t_k = 1/(k+1)$.

Conversely, $\sum_k t_k = \infty$ is *necessary* for this, for every proper closed convex f and x_0 (see (i)).



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arXiv:2609.35147, 2026 (Theorem 4.1)

Part 3

Proximal Point Method: Nonconvex Regime

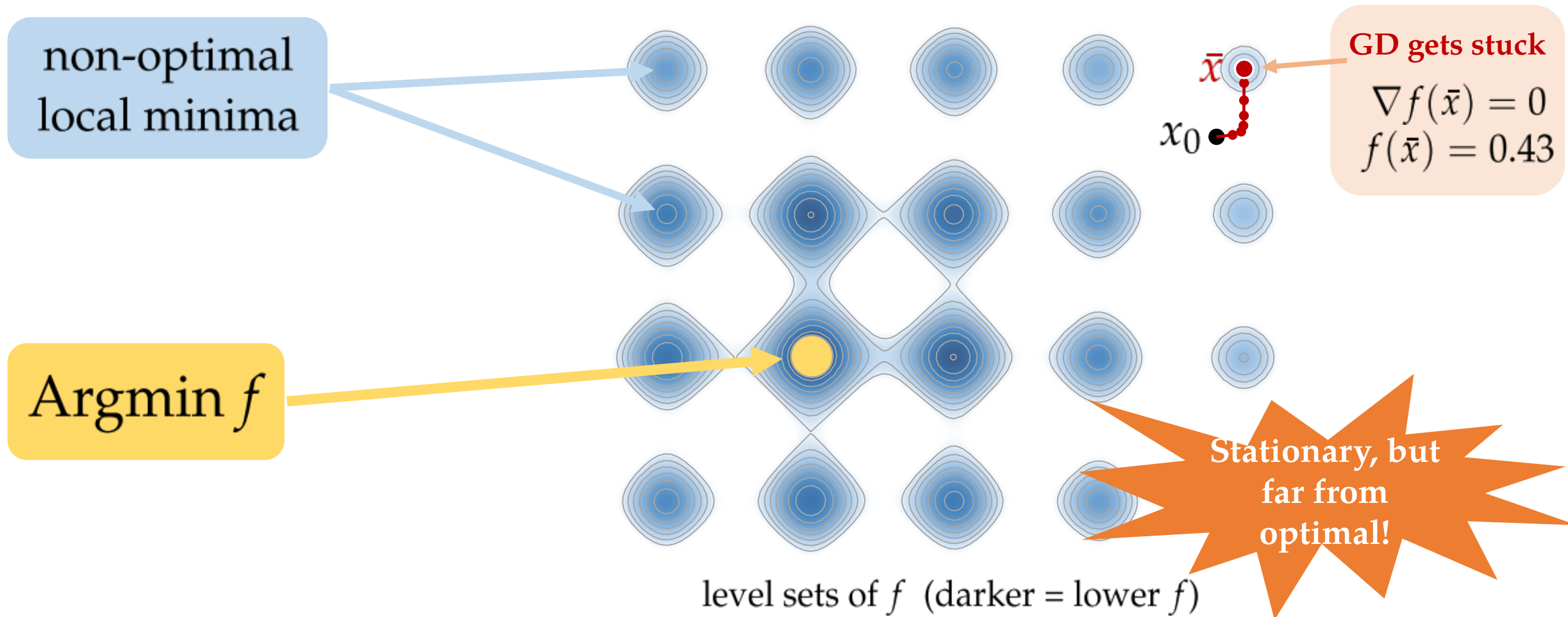


Kaja Grunkowska, Hanmin Li, Xun Qian, P.R.
Proximal alignment for global non-convex optimization
arXiv:2604.13483, 2026

2D Example 1: GD Gets Stuck

$$f(x) = \|x\|_2 + 10 \sin \zeta_1 + 10 \sin \zeta_2, \quad x = (\zeta_1, \zeta_2) \in \mathbb{R}^2$$

2D version of the 1D example $|x| + 10 \sin x$ (Grunkowska et al., 2026)



2D Example 1: BPM Jumps Over Local Minima

$$x_{k+1} \in \text{brox}_f^t(x_k) = \text{Argmin} \{f(y) : \|y - x_k\|_2 \leq t\}, \quad t = 2\pi$$

BPM function values

$$f(x_0) = 24.90$$

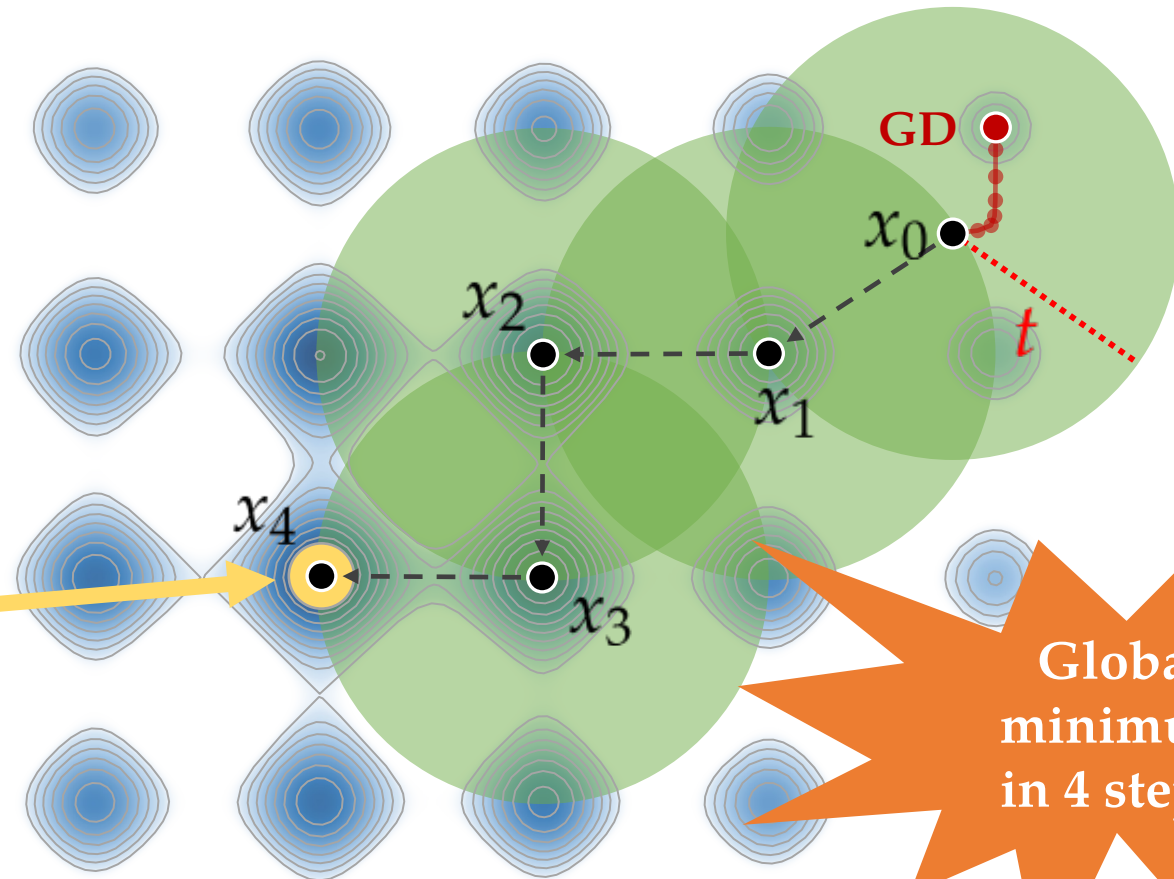
$$f(x_1) = -8.09$$

$$f(x_2) = -13.39$$

$$f(x_3) = -15.08$$

$$f(x_4) = -17.83 = f_\star$$

Argmin f



**Global
minimum
in 4 steps!**

level sets of f (darker = lower f)

2D Example 2: GD Gets Stuck

$$f(x) = \varphi(\|x\|_2) + \eta \text{ripple}(x), \quad \varphi(r) = r + 20 w(r) \sin^2(r/2)$$

radial analogue of the 1D example $|x| + 10 \sin x$; $w(r) = 1 - e^{-r^2/64}$; $0 \leq \text{ripple} \leq 1$, $\text{ripple}(0) = 0$; $\eta = 5$

60 non-optimal
local minima

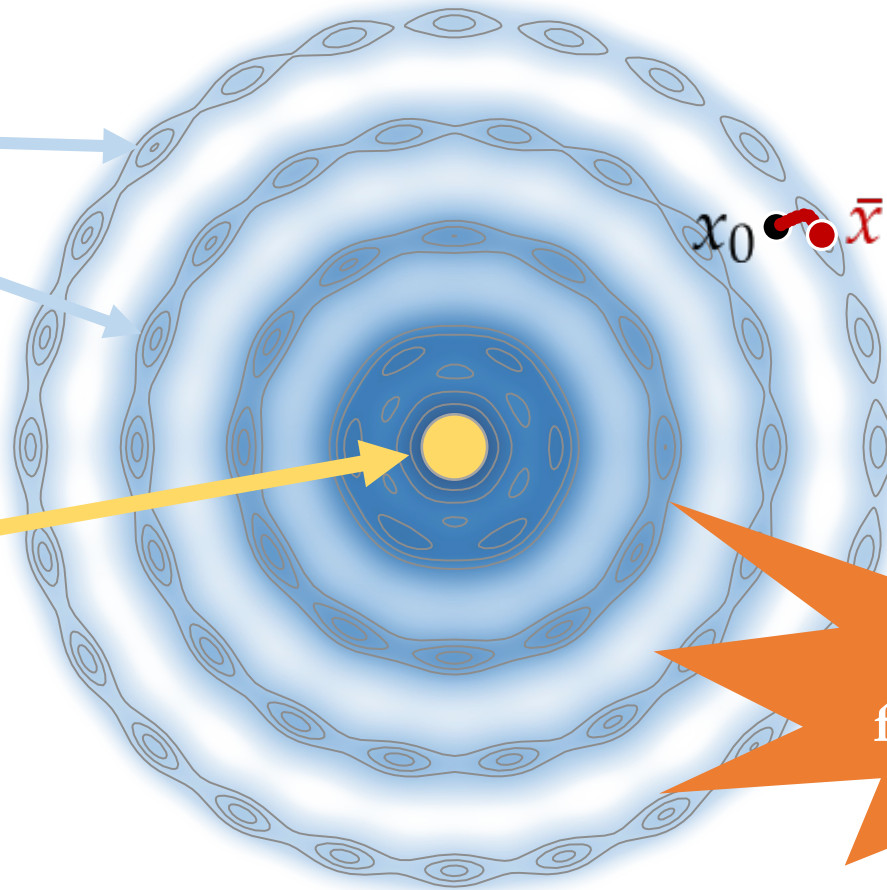
GD gets stuck

$$\begin{aligned} \nabla f(\bar{x}) &= 0 \\ f(\bar{x}) &= 26.30 \end{aligned}$$

level sets of f
(darker = lower f)

Argmin f

Stationary, but
far from optimal!



2D Example 2: BPM Jumps Over Local Minima

$$x_{k+1} \in \text{brox}_f^t(x_k) = \text{Argmin} \{f(y) : \|y - x_k\|_2 \leq t\}, \quad t = 2.5\pi$$

BPM function values

$$f(x_0) = 41.64$$

$$f(x_1) = 19.89$$

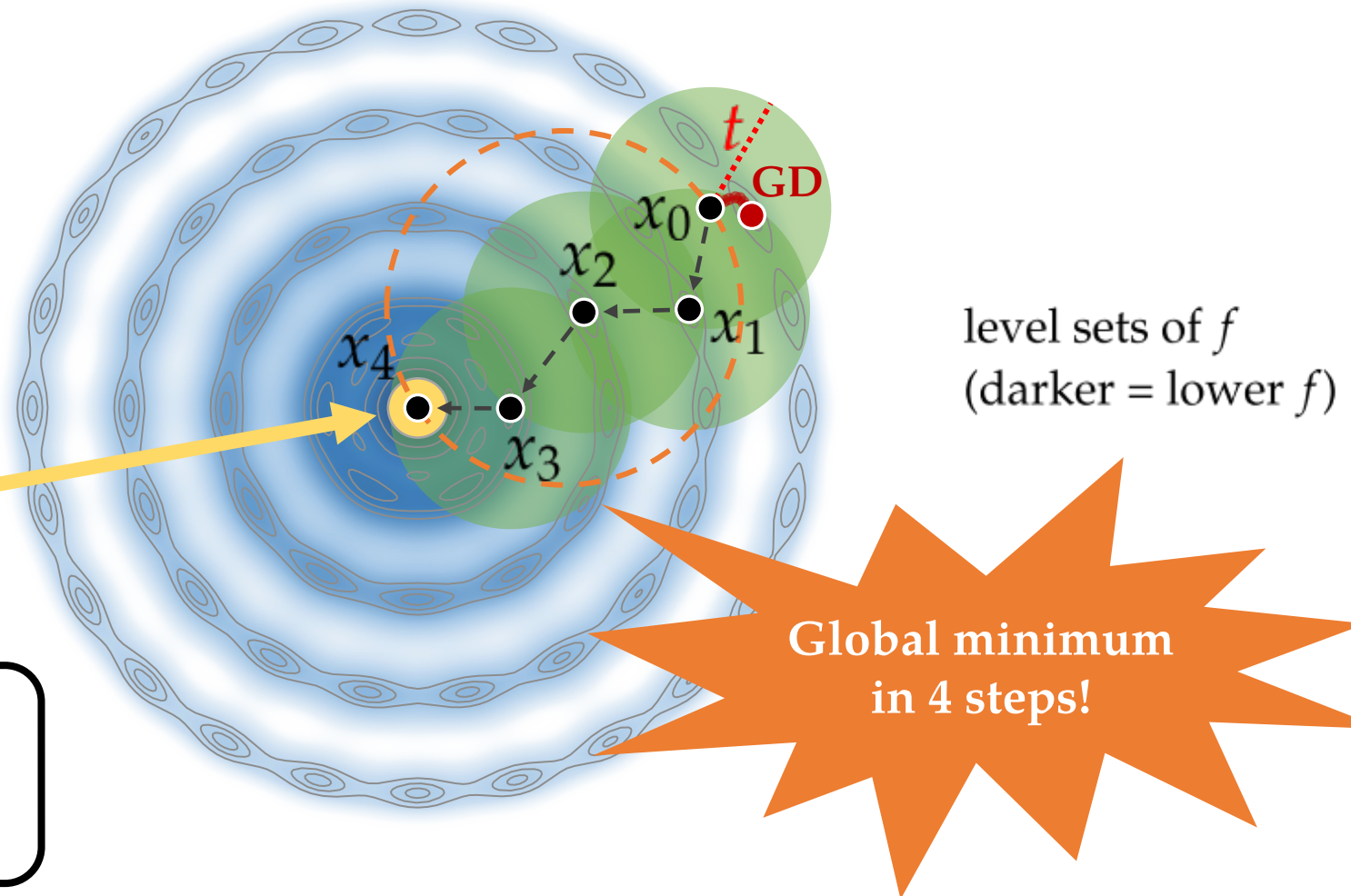
$$f(x_2) = 13.32$$

$$f(x_3) = 6.41$$

$$f(x_4) = 0.00 = f_\star$$

Argmin f

$f \in \mathcal{F}_I^{\text{BA}}(t) \Rightarrow$ Theorem 4.1 applies
Brox-aligned: each step lands in the ball
with diameter $[x_k, x_\star]$ (dashed: $k = 0$)



Brox-Aligned Nonconvex Functions

Broximal Alignment for Global Non-Convex Optimization

Table 1. Comparison of convex and non-convex function classes. **Non-Convex** = does not require f to be convex; **Non-Differentiable** = does not require f to be differentiable; **Stationary pts** = allows existence of stationary points that are not global minima; **Non-unique min** = allows existence of multiple global minima; **Disconnected** $\mathcal{X}_f$: the set of minimizers need not be connected.

Function class	Non-Convex	Non-Differentiable	Stationary pts	Non-unique min	Disconnected $\mathcal{X}_f$
Strongly convex	✗	✓	✗	✗	✗
Convex	✗	✓	✗	✓	✗
Pseudoconvex (Mangasarian, 1975)	✓	✗	✗	✓	✗
Quasiconvex (Arrow & Enthoven, 1961)	✓	✓	✓	✓	✗
Star-convex (Nesterov & Polyak, 2006)	✓	✗	✗	✓	✗
Quasar convex (Hardt et al., 2018)	✓	✗	✗	✓	✗
Aiming (Liu et al., 2023)	✓	✗	✗	✓	✗
Polyak-Łojasiewicz (Polyak, 1963; Łojasiewicz, 1963)	✓	✗	✗	✗	✗
Quadratic growth (Bonnans & Ioffe, 1995)	✓	✓	✓	✓	✓
$\mathcal{F}_{\mathbf{x}}^{\text{BA}}(t)$ (ours)	✓	✓	✓	✓	✓

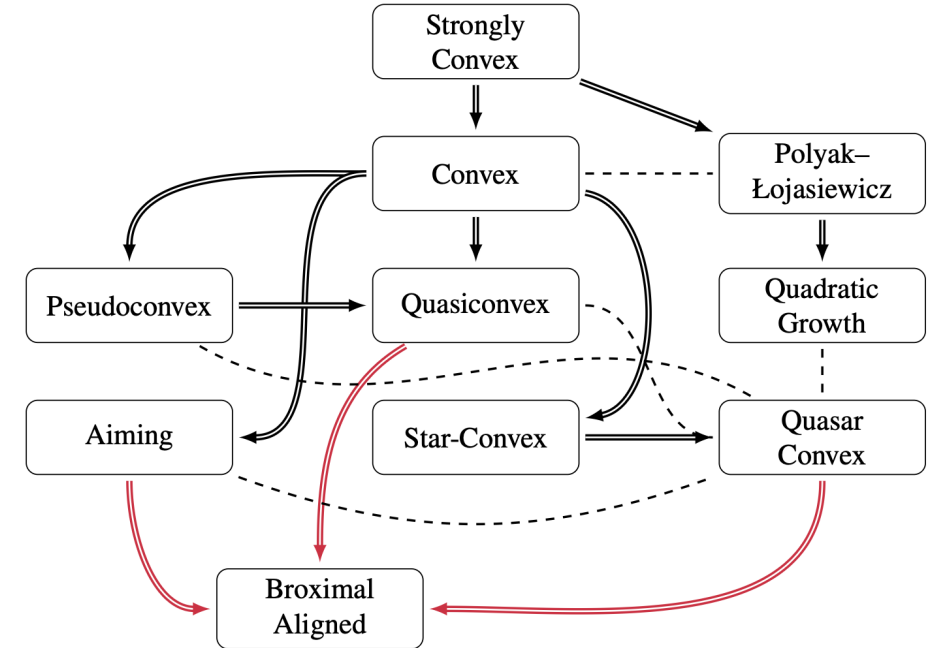


Figure 3. **Hierarchy of assumptions.** Solid arrows encode one-way implications, dashed lines denote mutual non-implications, and red arrows indicate conditions guaranteeing Broximal Alignment for any $t > 0$. For simplicity, we assume f is L -smooth (i.e., $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \forall x, y \in \mathbb{R}^d$) with domain $\text{dom } f = \mathbb{R}^d$. See the appendix and Karimi et al. (2016); Hinder et al. (2019); Khanh & Lara (2025) for proofs.



Kaja Grunkowska, Hanmin Li, Xun Qian, P.R.
Broximal alignment for global non-convex optimization
arXiv:2604.13483, 2026

Part 4
Proximal Point Method:
Practical Approximations

GD = Linearized BPM

$$x_{k+1} = \operatorname{argmin} \left\{ \underbrace{f(x_k) + \langle \nabla f(x_k), y - x_k \rangle}_{\text{linearization of } f \text{ at } x_k} : \|y - x_k\|_2 \leq t_k \right\} = x_k - \gamma \nabla f(x_k) \quad \text{if } t_k = \gamma \|\nabla f(x_k)\|_2$$

Example: $f(x) = \frac{1}{2} \langle a, x \rangle^2 + 2 \langle b, x \rangle^2$, $a = (\cos 20^\circ, \sin 20^\circ)$, $b = (-\sin 20^\circ, \cos 20^\circ)$

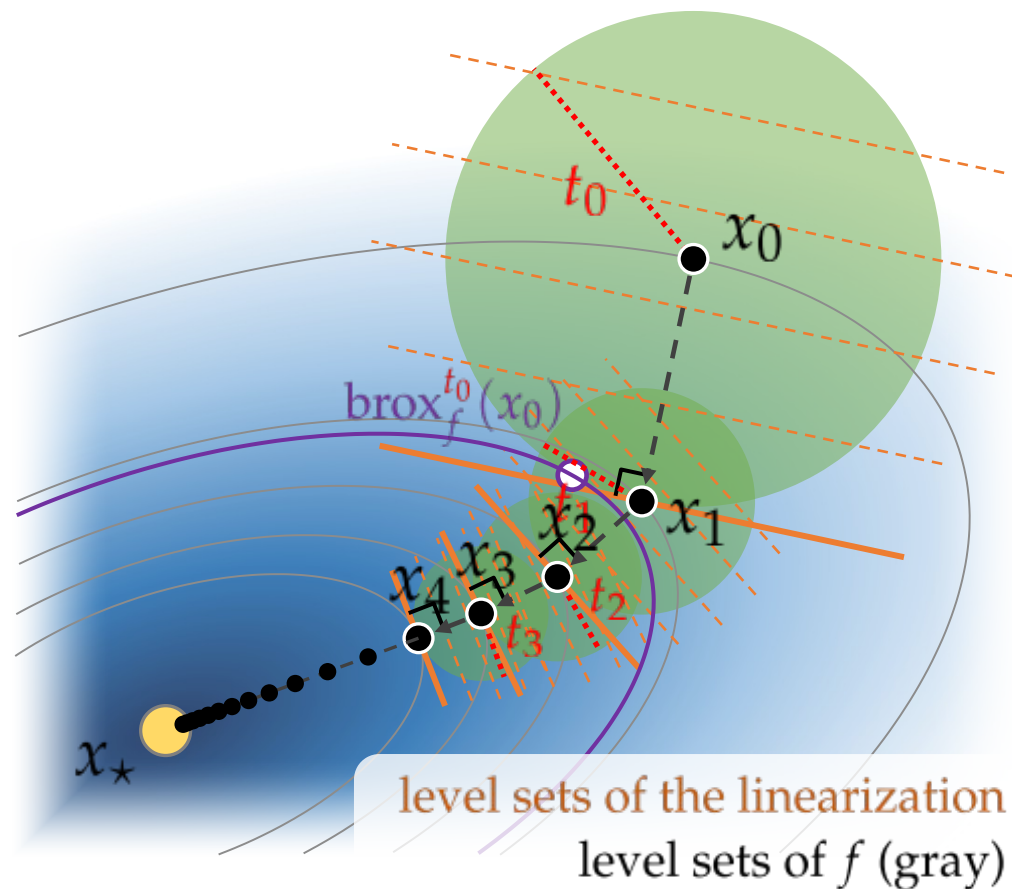
Linearize the objective inside the ball

linearize at x_k : $f(y) \approx f(x_k) + \langle \nabla f(x_k), y - x_k \rangle$

$$\operatorname{argmin} \{ \langle \nabla f(x_k), y \rangle : \|y - x_k\|_2 \leq t_k \} = x_k - t_k \frac{\nabla f(x_k)}{\|\nabla f(x_k)\|_2}$$

$$t_k = \gamma \|\nabla f(x_k)\|_2 \implies x_{k+1} = x_k - \gamma \nabla f(x_k) \quad (\text{GD})$$

GD = BPM on the linearization of f ,
with radius $t_k = \gamma \|\nabla f(x_k)\|_2$



SGD = Stochastically Linearized BPM

$$x_{k+1} = \operatorname{argmin} \left\{ \underbrace{f_{i_k}(x_k) + \langle \nabla f_{i_k}(x_k), y - x_k \rangle}_{\text{stochastic linearization of } f \text{ at } x_k} : \|y - x_k\|_2 \leq t_k \right\} = x_k - \gamma \nabla f_{i_k}(x_k) \quad \text{if } t_k = \gamma \|\nabla f_{i_k}(x_k)\|_2$$

Example: $f = \frac{1}{4} \sum_{i=1}^4 f_i$, $f_i(x) = \frac{1}{2} \langle a, x - c_i \rangle^2 + 2 \langle b, x - c_i \rangle^2$ (a, b as on the GD slide)

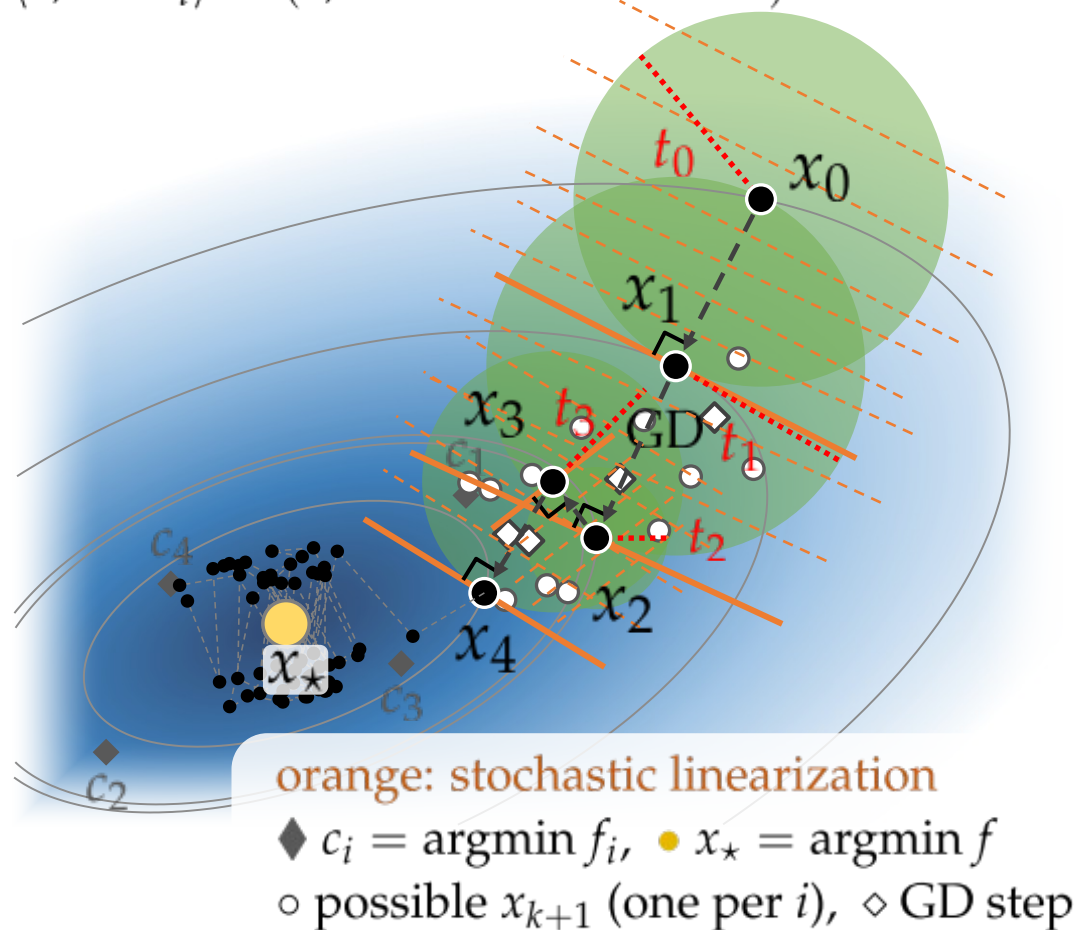
Linearize a random summand inside the ball

$$\ell_k(y) := f_{i_k}(x_k) + \langle \nabla f_{i_k}(x_k), y - x_k \rangle$$

$$\mathbb{E}_{i_k}[\ell_k(y)] = f(x_k) + \langle \nabla f(x_k), y - x_k \rangle \quad (\text{unbiased})$$

$$t_k = \gamma \|\nabla f_{i_k}(x_k)\|_2 \implies x_{k+1} = x_k - \gamma \nabla f_{i_k}(x_k) \quad (\text{SGD})$$

SGD = BPM on a random, unbiased linearization of f .
The radius does not vanish at x_* : SGD hovers around x_* .



SGD with Spectral-Norm Balls

$$X_{k+1} \in \underset{\substack{\text{stochastic linearization of } f \text{ at } X_k}}{\text{Argmin}} \left\{ \underbrace{f_{i_k}(X_k) + \langle G_k, Y - X_k \rangle}_{\text{stochastic linearization of } f \text{ at } X_k} : \|Y - X_k\|_{2 \rightarrow 2} \leq t_k \right\} \ni X_k - t_k U_k V_k^\top$$

Setting: $X \in \mathbb{R}^{m \times n}$, $\langle A, B \rangle = \text{tr}(A^\top B)$, $G_k := \nabla f_{i_k}(X_k) = U_k \Sigma_k V_k^\top$ (reduced SVD)

Linearize, then minimize over a spectral ball

$$\|D\|_F = \|\sigma(D)\|_2, \quad \|D\|_{2 \rightarrow 2} = \|\sigma(D)\|_\infty$$

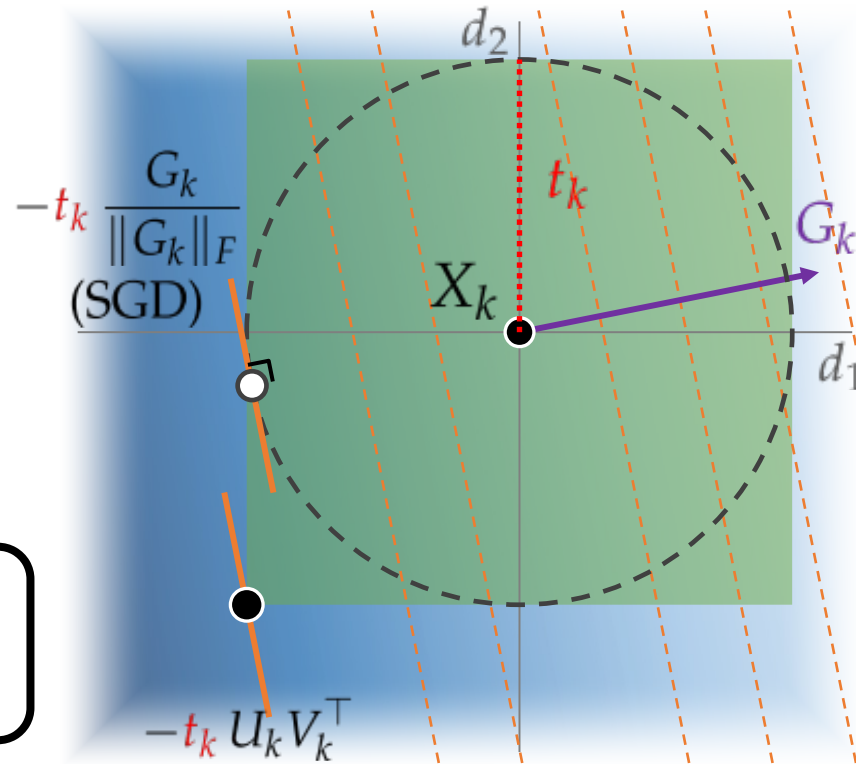
$$\|D\|_* = \|\sigma(D)\|_1 \quad (\text{nuclear norm})$$

Frobenius ball: $D = -t_k G_k / \|G_k\|_F$ (SGD)

spectral ball: $D = -t_k U_k V_k^\top$ (all $\sigma_i(D) = t_k$)

$$t_k = \gamma \|G_k\|_* \implies X_{k+1} = X_k - \gamma \|G_k\|_* U_k V_k^\top$$

Same recipe as SGD, only the ball changes:
the step is the orthogonalized gradient $-t_k U_k V_k^\top$.



green: $\|D\|_{2 \rightarrow 2} \leq t_k$
dashed: $\|D\|_F \leq t_k$
orange: $\langle G_k, \cdot \rangle$
 $\sigma(G_k) = (3, 0.6)$



Kaja Grunkowska, P.R.

Non-Euclidean broximal point method: a blueprint for geometry-aware optimization

arXiv:2510.00823, 2025

Muon: Adding Momentum and Newton–Schulz

$$M_k = \beta M_{k-1} + (1 - \beta) G_k, \quad X_{k+1} = X_k - \eta \text{NS}_5(M_k) \approx X_k - \eta U_k V_k^\top \quad (\text{Jordan et al., 2024})$$

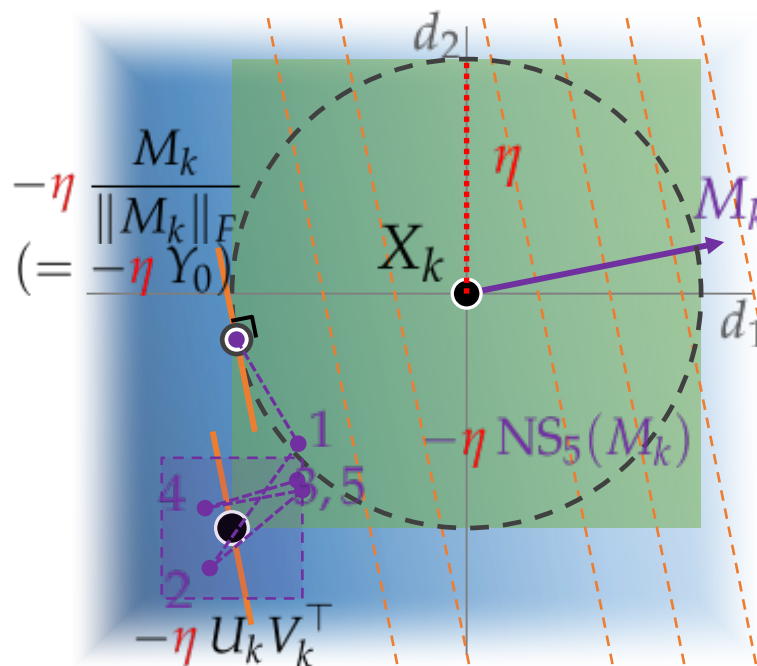
$$M_k = U_k \Sigma_k V_k^\top \text{ (reduced SVD)} : \quad X_k - \eta U_k V_k^\top \in \text{Argmin} \{ \langle M_k, Y - X_k \rangle : \|Y - X_k\|_{2 \rightarrow 2} \leq \eta \}$$

Spectral SGD (previous slide) + three changes

1. momentum: linearize with M_k , not G_k
2. fixed radius: $t_k \equiv \eta$ (fixed-radius theory)
3. inexact ball step: $U_k V_k^\top \approx \text{NS}_5(M_k)$

NS₅: 5 Newton–Schulz steps, $Y_0 \rightarrow Y_5 \approx U_k V_k^\top$

Muon $\approx$ BPM: *momentum* linearization,
spectral ball of *fixed* radius η ,
ball step computed *inexactly*.



green: $\|D\|_{2 \rightarrow 2} \leq \eta$
dashed: $\|D\|_F \leq \eta$
orange: $\langle M_k, \cdot \rangle$
purple: $-\eta Y_j$
 $\sigma(M_k) = (3, 0.6)$



Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, Jeremy Bernstein

Muon: An optimizer for hidden layers in neural networks

kellerjordan.github.io/posts/muon, 2024

BroxGD: Linearized BPM with Constraints

$$x_{k+1} \in \operatorname{Argmin} \left\{ \underbrace{f(x_k) + \langle \nabla f(x_k), y - x_k \rangle}_{\text{linearization of } f \text{ at } x_k} : \underbrace{y \in \mathcal{X}}_{\text{kept exactly}}, \|y - x_k\|_2 \leq t_k \right\}$$

Problem: $\min_{x \in \mathcal{X}} f(x)$, $\mathcal{X}$ closed, convex $\iff \min_x f(x) + \iota_{\mathcal{X}}(x)$: BPM with only f linearized

Between projected GD and Frank–Wolfe

local LMO on $\mathcal{X} \cap \mathcal{B}(x_k, t_k)$: no projection

ProjGD: soft ball, needs $\operatorname{Proj}_{\mathcal{X}}$

Frank–Wolfe: global LMO over $\mathcal{X}$

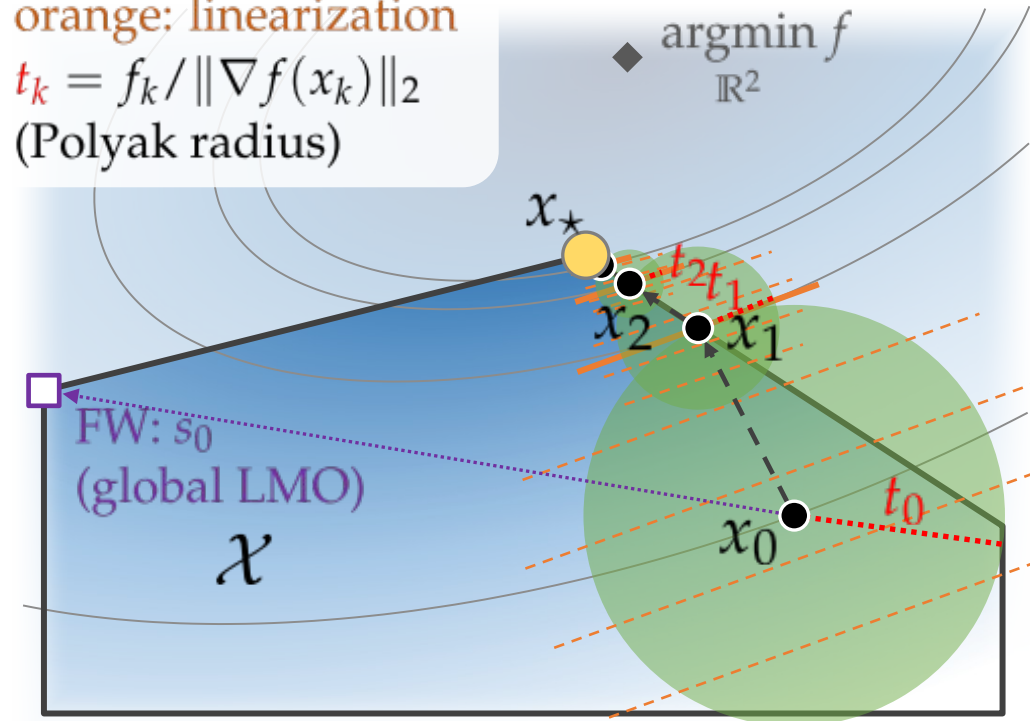
$$t_k \leq \frac{\langle \nabla f(x_k), x_k - x_{\star} \rangle}{\|\nabla f(x_k)\|_2} \implies d_{k+1}^2 \leq d_k^2 - t_k^2$$

BroxGD = BPM on $f + \iota_{\mathcal{X}}$ with only f linearized.

Smooth convex, $t_k = R/\sqrt{K}$: $f(\hat{x}_K) - f_{\star} \leq LR^2/(2K)$.

orange: linearization

$t_k = f_k / \|\nabla f(x_k)\|_2$
(Polyak radius)



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Broximal gradient descent: a projection-free sister of projected gradient descent

arXiv:2605.08850, 2026



P.R., Ammar Mahran

Local LMO is Secretly a Projection Method!

arXiv:2609.33383, 2026

Part 5

Proximal Point Method: Four Formulations



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A sharper theory of ball-proximal optimization: convergence and radius selection

arXiv:2609.35147, 2026

BPM = Projection onto Sublevel Sets

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = \text{Proj}_{C_k}(x_k), \quad C_k := \{y : f(y) \leq f(x_{k+1})\}$$

Sublevel projection

$$\min_{\|y-x_k\|_2 \leq t} f(y) = \alpha_k$$

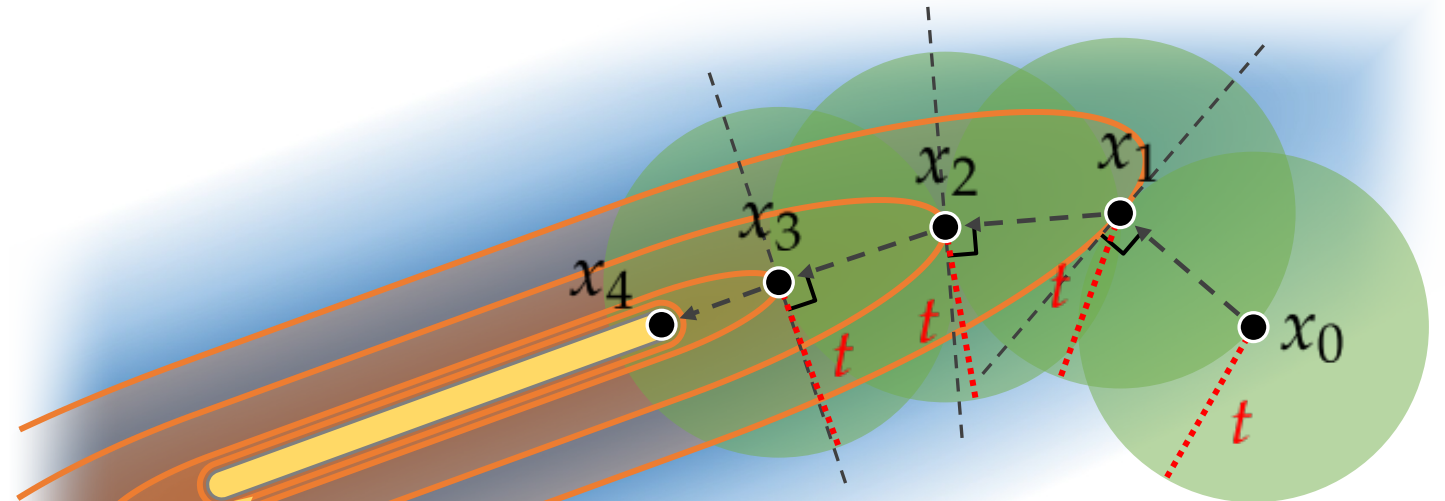
$$\iff \min_{f(y) \leq \alpha_k} \|y - x_k\|_2 = t$$

$$C_0 \supseteq C_1 \supseteq \dots \supseteq \text{Argmin } f$$

$$\implies d_{k+1}^2 \leq d_k^2 - t^2$$

$$\alpha_3 = 0 = f_* : x_4 = \text{Proj}_{\text{Argmin } f}(x_3)$$

Argmin f



sublevel set C_k (orange)
level sets of f (gray)
(darker = lower f)

BPM = Implicit Normalized GD

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = x_k - t \frac{\nabla f(x_{k+1})}{\|\nabla f(x_{k+1})\|_2}$$

Implicit (BPM) vs explicit

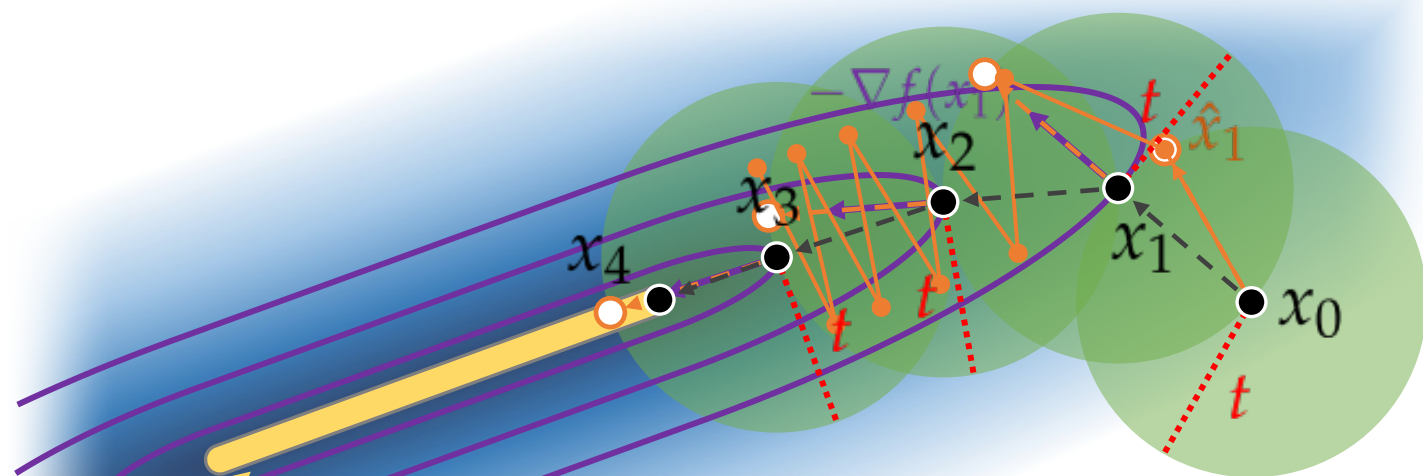
$$\hat{x}_{k+1} = x_k - t \frac{\nabla f(x_k)}{\|\nabla f(x_k)\|_2} \quad (\text{explicit})$$

$$x_{k+1} = \text{prox}_{\gamma_k f}(x_k), \quad \gamma_k = \frac{t}{\|\nabla f(x_{k+1})\|_2}$$

BPM: $f(x_4) = f_\star$

normalized GD: $f(\hat{x}_{10}) = 5.48$

Argmin f



explicit normalized GD (orange)
level set through x_{k+1} (purple)
(darker = lower f)

BPM = Normalized GD on the Ball Envelope

$$x_{k+1} \in \text{brox}_f^t(x_k) \implies x_{k+1} = x_k - t \frac{\nabla N_f^t(x_k)}{\|\nabla N_f^t(x_k)\|_2}$$

Ball envelope

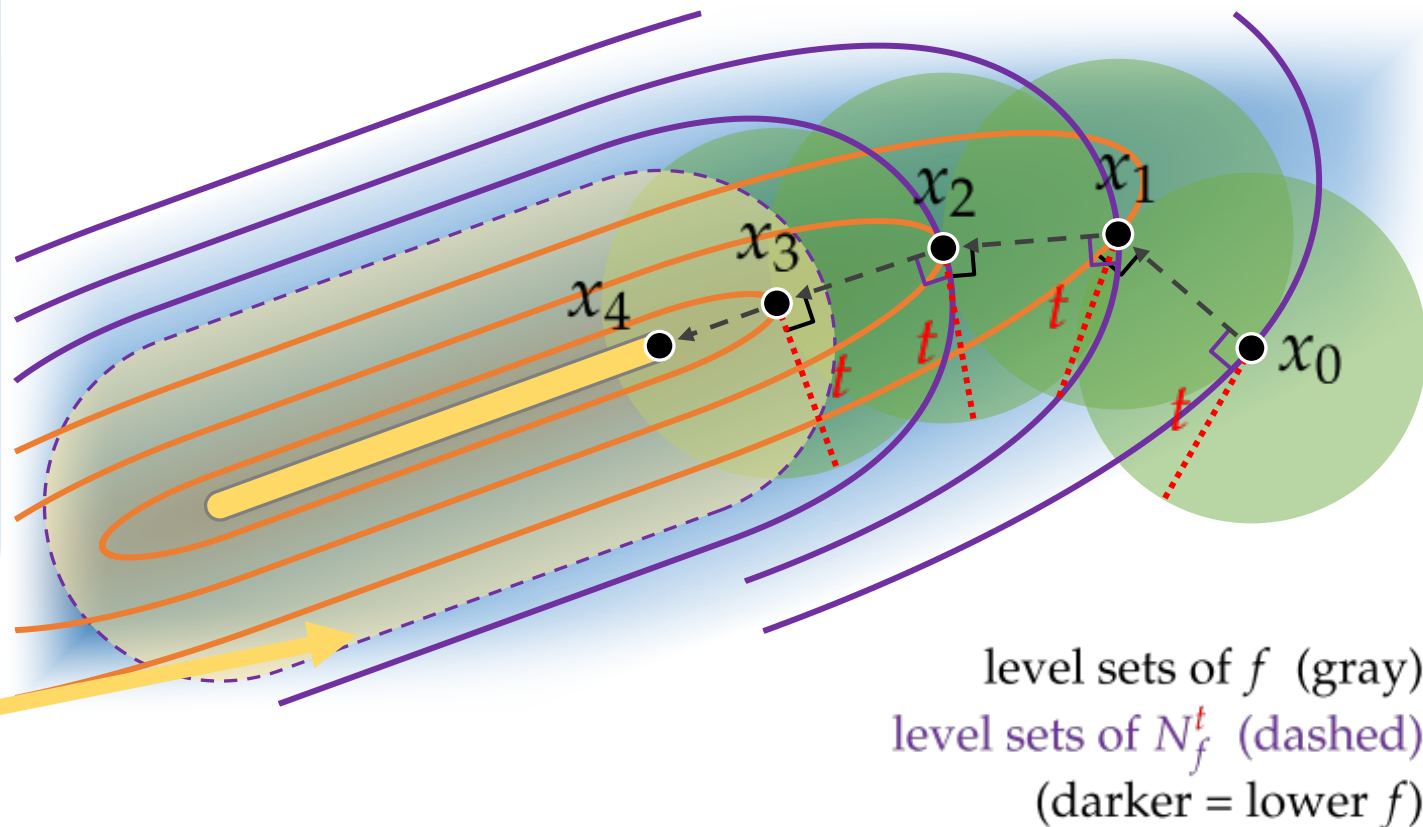
$$N_f^t(x) := \min_{\|y-x\|_2 \leq t} f(y)$$

$$\{N_f^t \leq c\} = \{f \leq c\} + \mathcal{B}(0, t)$$

$$N_f^t(x_k) = f(x_{k+1}), \quad \nabla N_f^t(x_k) = \nabla f(x_{k+1})$$

$$N_f^t(x_3) = 0 \implies x_4 \in \text{Argmin } f$$

$$\text{Argmin } N_f^t = \text{Argmin } f + \mathcal{B}(0, t)$$



BPM = Implicit Regularized Newton

averaged Hessian

$$\bar{H}_k := \int_0^1 \nabla^2 f(x_k + \tau(x_{k+1} - x_k)) d\tau$$

$$x_{k+1} = x_k - \left(\bar{H}_k + \underbrace{\lambda_k I}_{\text{regularization}} \right)^{-1} \nabla f(x_k)$$

Example: $f(x) = \frac{1}{3}(|\langle a, x \rangle|^3 + 8|\langle b, x \rangle|^3)$, $t = 0.6$

Why? (convex, twice-differentiable objective)

optimality: $\nabla f(x_{k+1}) + \lambda_k s_k = 0$, $\lambda_k = \|\nabla f(x_{k+1})\|_2 / t$

mean value: $\nabla f(x_{k+1}) - \nabla f(x_k) = \bar{H}_k s_k$, $s_k := x_{k+1} - x_k$

$\Rightarrow (\bar{H}_k + \lambda_k I) s_k = -\nabla f(x_k)$ (implicit: $\bar{H}_k, \lambda_k$ depend on x_{k+1})

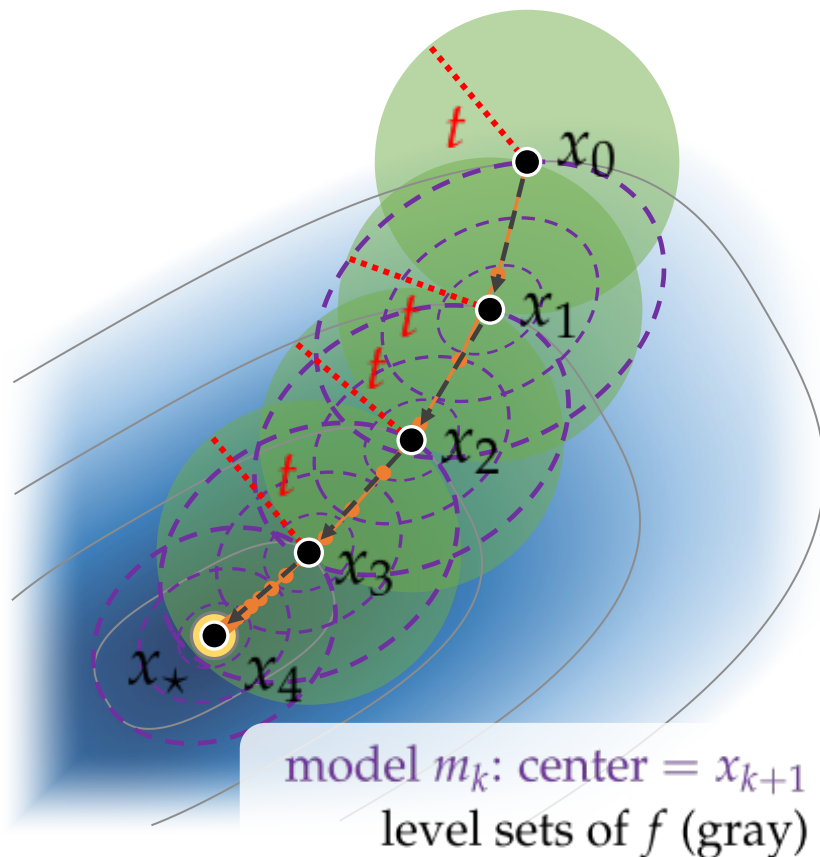
Computable: replace $\bar{H}_k \rightarrow \nabla^2 f(x_k)$ and $\lambda_k \rightarrow \sqrt{\sigma \|\nabla f(x_k)\|_2}$
 $\Rightarrow$ regularized Newton (Mishchenko, 2023): orange, $\sigma = 8$



P.R.

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Approximate Broximal Point Methods

Manuscript, 2026





Conclusions

The BPM Family Tree

BPM

$$x_{k+1} \in \underset{y}{\operatorname{Argmin}} f(y)$$

$$\text{s.t. } \|y - x_k\| \leq t_k$$

linearize f

Gradient descent

linearize a random summand f_i

SGD

linearize, spectral-norm ball, momentum

Spectral SGD, Muon

linearize f , keep $\mathcal{X}$ exactly

BroxGD

$$\nabla f(x_{k+1}) \rightarrow \nabla f(x_k)$$

Normalized GD

$$\bar{H}_k \rightarrow \nabla^2 f(x_k), \lambda_k \rightarrow \sqrt{\sigma \|\nabla f(x_k)\|_2}$$

Regularized Newton

ball $\rightarrow$ penalty $\|y - x_k\|_2^2 / (2\gamma)$

Proximal point method

Conclusions

$$\text{BPM: } x_{k+1} \in \text{Argmin} \{f(y) : \|y - x_k\| \leq t_k\}$$

Part 1 Convex Regime

any norm ball ($\ell_2, \ell_1, \ell_\infty$): exact solution in a few steps
LP $\rightarrow$ a few small LPs, each using only a few constraints

Part 2 Convergence Theory

$d_{k+1}^2 \leq d_k^2 - t^2$: exact solution after $\lceil d_0^2/t^2 \rceil$ steps; no condition number
any radii: $\sum_k t_k = \infty \Rightarrow f(x_k) \downarrow \inf f$

Part 3 Nonconvex Regime

BPM jumps over non-optimal local minima where GD gets stuck
brox-aligned f : Theorem 4.1 still applies $\Rightarrow$ global minimum

Part 4 Practical Approximations

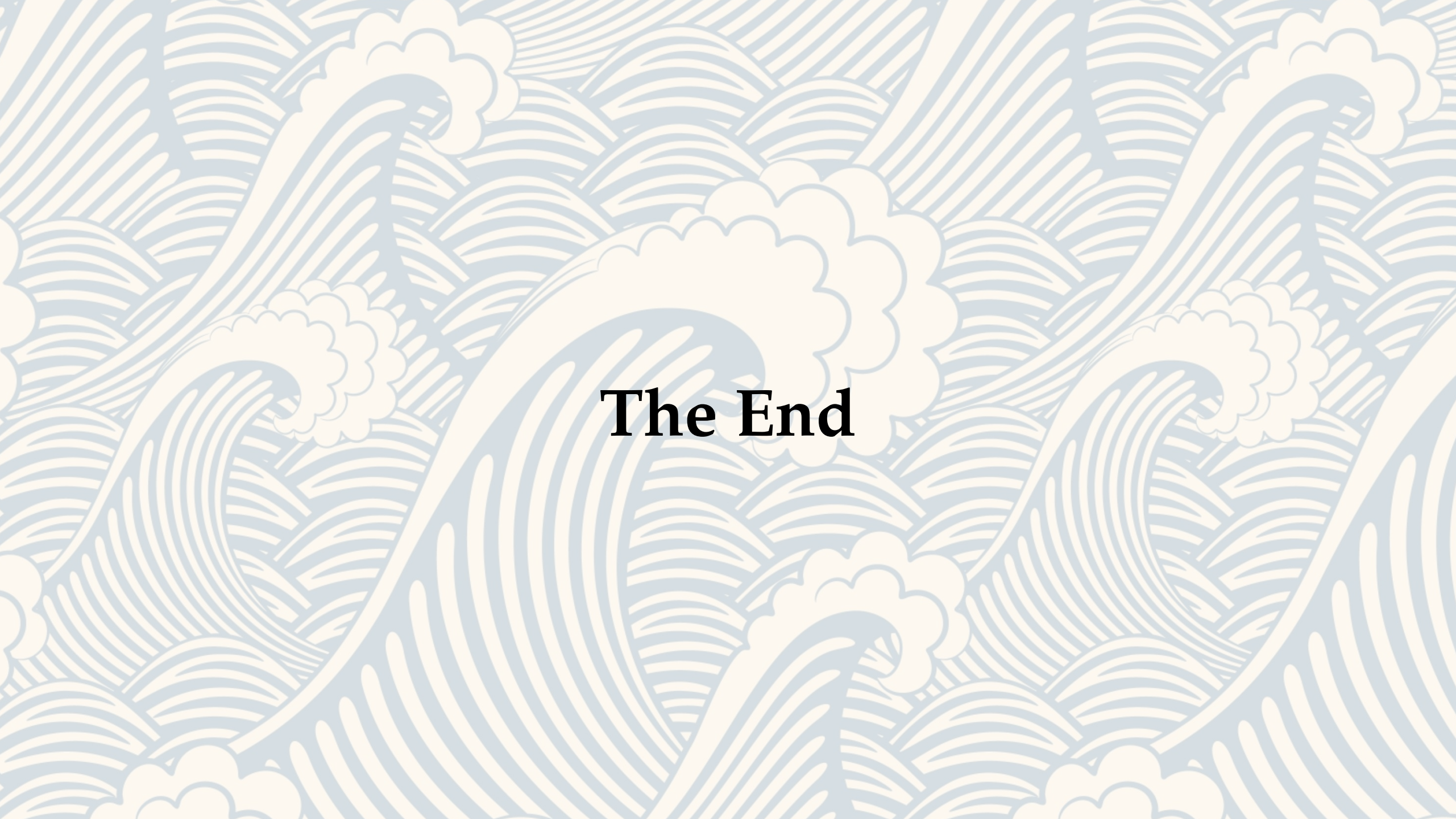
linearize f inside the ball: GD, SGD, spectral SGD, Muon, BroxGD
the ball sets the geometry: spectral $\rightarrow$ Muon, ball $\cap \mathcal{X} \rightarrow$ BroxGD

Part 5 Four Formulations

BPM = projection onto sublevel sets = implicit normalized GD
= normalized GD on the ball envelope = implicit regularized Newton

How to think about optimization: *minimize f over a ball, step after step.*

Practical methods approximate this step: a ball + an approximation of f + a radius rule.



The End